



CHAPTER 12

AIR CONDITIONING AND HEAT PUMPS

LEARNING OBJECTIVES

By the end of this chapter you should be able to:

- define in one sentence each the function of the compressor, condenser, evaporator and expansion device
- list ten factors that affect how much air conditioning is needed
- explain the implications of an undersized air conditioning system
- explain the implications of an oversized air conditioning system
- describe the location of the air conditioning compressor
- describe in one sentence the function of a crankcase heater
- list eight common compressor problems
- describe the location and function of the condenser fan and the evaporator fan
- list four common condenser fan problems
- list seven common evaporator fan problems
- list eight common duct problems
- list seven thermostat problems
- give the normal life expectancy for conventional air conditioner compressors
- list nine tricks for identifying heat pumps
- list seven common heat pump problems

INTRODUCTION

Central air conditioning is considered a luxury rather than an essential in most parts of the United States. Air conditioning systems are more common in the southern areas, and are more common in humid areas than dry areas. There are probably more air conditioners per capita in Florida than in California, for example. You probably know how common central air conditioning systems are in your area.

Scope of Inspection

In this chapter we'll look at split-system air-cooled central air conditioning, since it's the most common system in homes. We inspect permanently **installed** systems with or without ducts. We don't inspect window air conditioners or other systems that can be removed without tools and that plug into 120-volt convenience receptacles. We'll conclude this chapter with a look at heat pumps, a kind of air-conditioner-in-reverse.

12.1 AIR CONDITIONING CAPACITY

Before looking at the specific components of air conditioners and what can go wrong with them, let's consider a general issue which concerns many homeowners: Is the system capacity sufficient to keep the house comfortable during hot weather?

Most home inspectors give their clients some indication as to whether the air conditioning system is sized properly even though it's not required by the Standards. Let's look at the cooling capacity.

Factors Affecting Cooling Load

The amount of cooling required depends on a large number of factors. These include the outdoor temperature; the outdoor humidity; the level of insulation in the house; the amount of air leakage in the house; the amount of southern, east and west facing glass in the house; whether this glass is single-, double- or triple-glazed; whether the glass is a low emissivity glass or gas-filled; and whether window treatments (curtains or blinds) are kept closed or open. Other factors include the amount of shading from trees, roof overhangs, awnings, or buildings and how much heat is generated in the house by the people and equipment inside.

Guidelines

Despite all these variables, most people like to have guidelines. Home inspectors are no exception. In the southern United States, 450 to 700 square feet of floor area per ton of cooling is considered appropriate. In the northern United States, 700 to 1,000 square feet per ton may be adequate. Speak to air conditioning installers and other inspectors in your area to find the appropriate range for your area. (These guidelines assume 8-foot ceilings.)

Duct Capacity Problems

The capacity of the equipment is only one part of the equation. Many air conditioners that under-perform are a result of a duct system incapable of circulating the conditioned air adequately through the system. This is particularly true where air conditioning has been added to a house with ducts that were designed for a heating system only.

Moving Heavy Air with More Obstructions

Adding central air conditioning to an existing furnace system may lead to inadequate air distribution for several reasons. Firstly, the evaporator coil presents an additional obstruction to airflow and reduces the rate of air movement through the system. Secondly, during the cooling season, we are trying to move air that is at 55°F rather than air that is at 140°F (which is what we see in the heating season). The cooler air is more dense (heavier) and more difficult to move through the ducts. We also have to move more air since the difference in temperature between the con-

ditioned air and the room air (about 15°F to 20°F) is less than with a conventional oil or gas furnace (60°F to 70°F).

Air conditioning systems typically move 400 to 450 cubic feet of air per minute per ton through a system. Heating systems only need to move about half this much.

Many air conditioning manufacturers and installers recommend slightly undersizing an air conditioning system, rather than oversizing. Either can be a problem.

Better to Undersize

12.1.1 Undersized

IMPLICATIONS

During moderate weather, the air conditioner may function adequately, but during hot weather, the air conditioner may not be able to achieve a 15°F to 20°F temperature differential between indoors and outdoors.

STRATEGY

The first step is to determine the size of the air conditioning system. This can often be done by reading the model number on the data plate. This is typically located on the outdoor (condenser) unit. The size may be recorded in thousands of BTUs per hour, or in the number of tons.

Preston's Guide

Sometimes it is difficult to translate a model number into a system capacity. **Preston's Guide** is an excellent reference guide with the model, serial numbers and SEER (Seasonal Energy Efficiency Ratings) of many residential air conditioning systems.

Guessing the Size

If the size cannot be determined from the model number on the data plate, the size can be approximated from the Rated Load Amperage (RLA) on the data plate (Figure 12.1). A typical reciprocating compressor will be rated at 6 amps to 8 amps per ton of cooling. The newer high-efficiency units and scroll compressors will draw less electrical current, more like 5 amps per ton. Be clear that this is an approximation only if you report the capacity based on current.

Use House Square Footage

The next step is to roughly calculate the above-grade square footage of the home. Divide the square footage into the number of tons and determine the number of square feet per ton.

If the number of square feet per ton exceeds the ranges we discussed, it is probably best to describe this as marginal or suspect capacity and to recommend further investigation. There may be factors that cause the guidelines not to apply.

Guidelines

It's also possible to find a system that seems to be just fine with respect to capacity using your guideline and yet it isn't really big enough. When considering the square

FIGURE 12.1 Guessing the Size

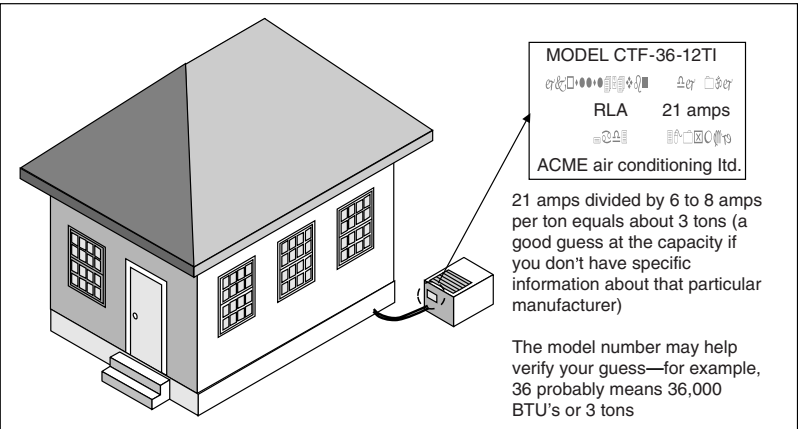
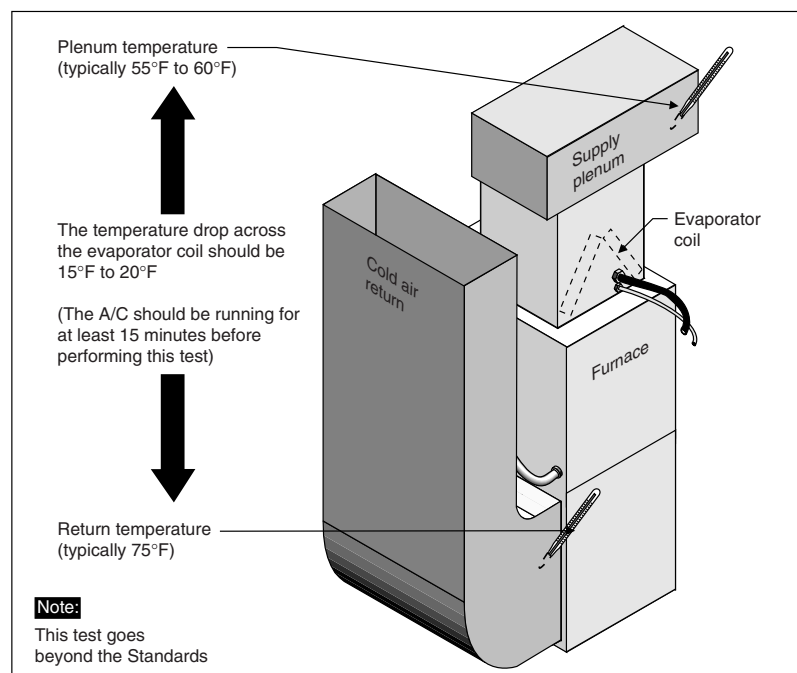


FIGURE 12.2 Measure Temperature Drop across Inside Cold

Measure Temperature Drop Across Indoor Coil

footage of the house, the basement is not usually considered. However, if the basement has a walk-out with a large glass surface facing south, east or west, the air conditioning load may be far greater than contemplated. Lots of skylights change things too.

If the system is adequately sized and is working properly, the air temperature entering the evaporator coil will be whatever the room temperature is (Figure 12.2). Let's say it's 75°F. The air coming off the coil should be 14°F to 22°F cooler (some say 15°F to 20°F). If the inlet temperature is 75°F, the air coming off should be 55°F to 60°F. This can be measured with a thermometer with a sharp probe that is pushed into a joint or hole in the supply plenum immediately downstream of (after) the evaporator coil.

If the temperature drop is different, the problem may be size-related or may indicate a need for servicing. This test should be compared with your approximation of the size of the air conditioner, based on the number of square feet per ton. Make sure the temperature drop is measured after the system has established equilibrium. The unit should run for at least 15 minutes before checking the temperature split. Measuring this temperature split is beyond the Standards but many inspectors do it.

12.1.2 Oversized

An oversized air conditioner is susceptible to short cycling, inadequate dehumidification and large temperature variations in the house.

Oversized units will have a shortened life expectancy and will provide a less comfortable environment. The largest comfort issue is the lack of dehumidification. Because the temperature drops rapidly with an oversized unit, there is not enough air movement across the coil to extract the water from the house air. This results in

IMPLICATION

STRATEGY

a house that is cool, but with a humid, swamp-like environment. Since compressors experience most damage on start-up, short cycles also mean more start-ups and a shorter life.

Other than the rough guideline test, it is difficult to know whether and how much the unit is oversized. Some utilities indicate that a unit may be as much as 25 percent oversized without adverse effect. The temptation to oversize may become irresistible for installers with respect to heat pumps. Since heat pumps have to deal with a much larger temperature differential from outside to inside, the tendency is to make the heat pump large enough to meet the heating demand. This makes it too large for the cooling load. There are some strategies to address this problem, but within this context, we are watching for oversized cooling units.

One way inspectors identify an oversized air conditioner is by sensing the cold damp environment when walking into a house. Also, an air conditioner that short cycles (turns on and off every five minutes) is a suggestion that the unit may be oversized.

Two surveys have shown that one third to one half of all residential air conditioning systems are oversized.

12.2 COMPRESSOR

The compressor is the heart of the air conditioning system. It is the pump that drives the refrigerant through the system.

Location

On conventional split-system systems, the compressor is in the condenser cabinet outdoors or in the attic. It is typically a large, black, hermetically sealed cylinder.

The motor and compressor operate in a refrigerant-filled environment. Opening the condenser cabinet to get at the compressor is going beyond the Standards.

Condensers Hung on Houses

Most condensers sit on the ground but some hang on brackets attached to the building wall. This keeps them out of the dirt and eliminates settling problems. Some claim that this transfers noise and vibration into the house.

Low Temperature Limitations

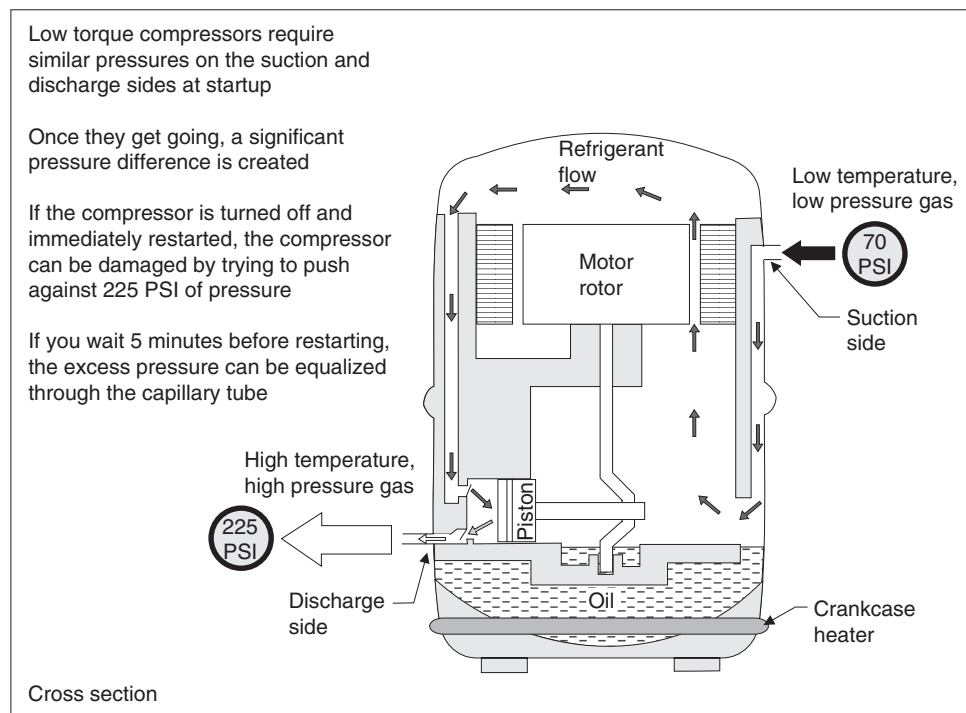
Compressors shouldn't be tested when the outdoor temperature is below 65°F or when electrical power has been on for less than 12 to 24 hours. Damage can be done to the compressor. The refrigerant in the compressor may condense and mix with the lubricating oil in the base of the compressor. This mix does not provide good lubrication and the compressor may seize.

Sump or Crankcase Heater

There is a sump heater (crankcase heater) on many air conditioning systems that keeps the oil at the base of the compressor warm enough to boil off the refrigerant. The heater usually is on whenever the air conditioning system is powered, but is off when the unit is shut down for the heating season. It's a waste of electricity to keep this heater on year round. The heater may take 12 to 24 hours to warm the oil to the point where all the refrigerant is boiled off. That's why you can't just turn the power on and start up the air conditioner at the beginning of the cooling season.

Delayed Start-Up

Compressors that use capillary tubes as expansion devices are typically low-torque units. This means that on start-up, they expect to see equal pressure on the suction and discharge side of the compressor. The capillary tube allows this equalization of pressure but it takes a little time. When the compressor shuts off, it has a high pressure on the discharge side (as high as 275 psi with some refrigerants) and a low pressure on the suction side (typically 70 psi).

FIGURE 12.3 Wait Before Restarting a Compressor*Compressor Damage*

If the compressor is asked to start again immediately, it will be trying to push against a high pressure (see Figure 12.3). This can damage the compressor. As a result, most air conditioning systems have a five-minute delay built in that prevents the compressor from coming on just after it has stopped.

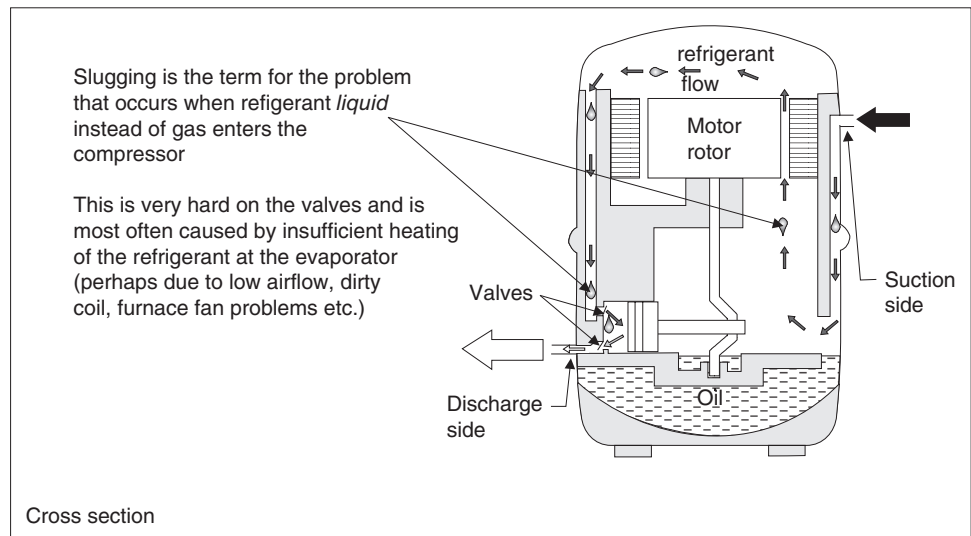
Conditions

Common compressor problems include the following:

1. Excess noise/vibration
2. Short cycling or running continuously
3. Out of level
4. Wrong fuse or breaker size
5. Electric wires too small
6. Missing electrical shutoff
7. Inoperative
8. Inadequate cooling

12.2.1 Excess Noise/Vibration

Compressors are not silent when operating. They should run with a monotonous drone. Knocking sounds coming from the compressor are cause for concern. A hum with no background compressor noise may indicate that the compressor is inoperative. No noise at all also indicates that the compressor isn't working.

FIGURE 12.4 Slugging*Normal Noises*

Scroll-type compressors have a different sound than reciprocating compressors. They have a higher pitched whine. It takes some experience to determine what is typical compressor noise. Some compressors are noisy when new and will break in over time.

Vibration

Some vibration is common with any compressor. Compressors are mounted on rubber feet to isolate this vibration. Excessive vibration, often accompanied by unusual noise, is a sign of severe problems with the system.

IMPLICATIONS

Unusual noises often indicate imminent failure of the compressor. Excess vibration leads to joint failure in the refrigerant line connections. This will allow refrigerant loss, which leads to compressor burnout. Internal damage may also be done to the compressor.

STRATEGY

Compressor noise can be heard best if you are close to the compressor. Make sure you don't confuse the sound of the outdoor coil fan with compressor noise.

Listening with a Screwdriver

Some inspectors remove the outdoor cover and press the tip of a screwdriver against the shell of the compressor and the base of the screwdriver against the ear drum. This will transmit sounds to your ear without background noise from the fan, allowing a more accurate assessment.

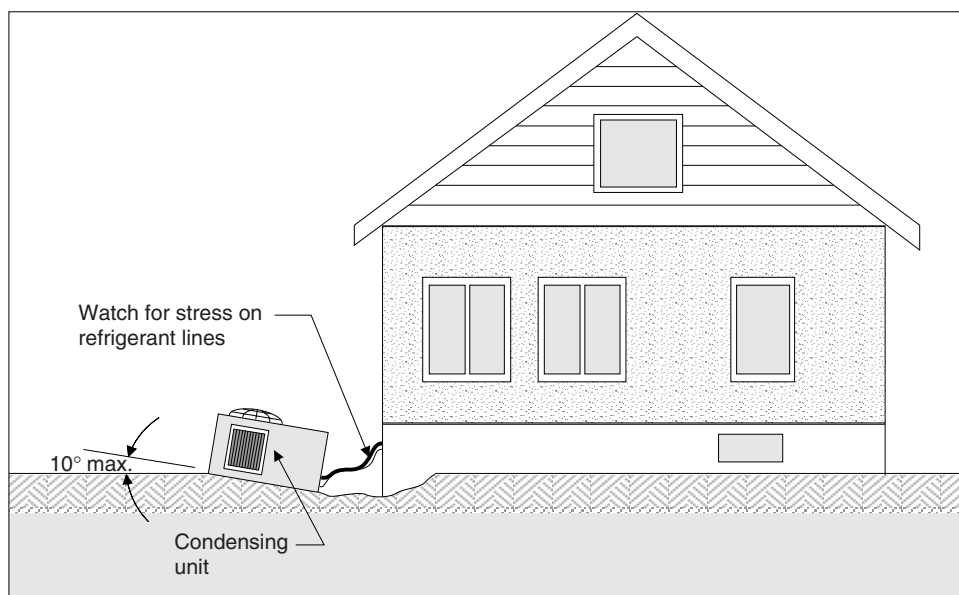
Slugging

Compressors that are slugging may be noisy intermittently. Slugging is the introduction of liquid to the intake side of the compressor (see Figure 12.4). Compressors are not intended to work on liquids. They expect to see a gas. The liquid may be oil or liquid refrigerant. In either case, slugging is very hard on a compressor.

When the compressor is running, look for evidence of vibration. This can sometimes be seen through the condenser cover or through the coil. Intermittent vibration or vibration which is readily visible may indicate problems. Look at the rubber mounts to make sure they are secure.

Don't worry about vibration on start-up or shutdown. This is typical. However, a broken mount will show up clearly at start-up and shutdown.

Where noise levels or vibrations are unusual, recommend further investigation.

FIGURE 12.5 Condensing Unit Out of Level

12.2.2 Short Cycling or Running Continuously

IMPLICATIONS

Compressors that exhibit short cycling or run continuously may result in ineffective cooling and shortened life expectancy of the compressor.

STRATEGY

Watch for an air conditioner that never shuts off on a mild day (on a very hot day, it's likely to run continuously) or comes on and off every five minutes. Troubleshooting is beyond our scope, but recommend that the unit be serviced.

12.2.3 Out of Level

The condensing units should be within approximately 10 degrees of level (see Figure 12.5).

IMPLICATIONS

The compressor may not be properly lubricated if the unit is not level.

The other implication of a compressor out of level is slugging. This is very hard on the compressor.

Oil traveling with the refrigerant through the tubing may become trapped if the unit is out of level. This reduces the lubricant available to the compressor and may cause compressor failure.

Another implication of condensing units being out of level is the refrigerant lines breaking as a result of the stresses placed on lines by the unit.

STRATEGY

Look at the condenser unit to see that it is approximately level. No tools are needed. While you are at it, check that the unit is stable and doesn't wobble. Watch for loose or rusted legs.

12.2.4 Wrong Fuse or Breaker Size

The fuse or breaker must be the right size to adequately protect the air conditioner and its wiring. The maximum fuse size is usually on the data plate. Ensure the size is correct.

Damage to the condenser unit, and possibly a fire, can result from the wrong fuse or breaker size.

A visual check of the data plate specification against the overcurrent protection device (fuse or breaker) is all that is needed.

If the data plate gives only a fuse size rating, only fuses should be used. Circuit breakers shouldn't be used on this system. If the data plate says "HACR Breaker" this means a **Heating, Air Conditioning, Refrigeration** rated breaker. Most breakers sold today are HACR rated, but look for the rating on the breaker.

If the data plate is missing or partly illegible, you can approximate the correct fuse or breaker size by using 125 percent of the total of the compressor RLA and the condenser fan FLA (full-load-amperage).

IMPLICATIONS
STRATEGY

No Rating Found

12.2.5 Electric Wires Too Small

The wire carrying electricity to the condenser unit may be too small.

Implications include:

- overheating of the wire and a possible fire
- excess voltage drop
- an inoperative or poor performing compressor

Check the data plate to determine the **minimum circuit ampacity**. Make sure the wires serving the condenser unit are large enough to carry this current.

If the data plate is missing or illegible, you can approximate the minimum wire rating by multiplying the compressor RLA plus the condenser fan FLA (full load amperage) by 125 percent.

IMPLICATIONS

STRATEGY

No Data

12.2.6 Missing Electrical Shutoff

In much of the United States, an outdoor electrical disconnecting means is required within sight of and readily accessible to the condenser unit. Check with your local authorities to determine whether this is needed in your area.

A missing electrical shutoff means the unit cannot be serviced conveniently.

Look for an accessible disconnecting means (switch) for the air conditioner close to the condenser unit. The switch should not be behind or right above the condenser unit because it may not be readily accessible.

IMPLICATION

STRATEGY

12.2.7 Inoperative

Obviously, an inoperative compressor means no air conditioning.

If the compressor doesn't operate when you turn the thermostat down, check first to make sure the fuse or breaker is active. Make sure the thermostat is set to the "Cool" position. If the thermostat is a mercury bulb type, make sure the switch

IMPLICATION

STRATEGY

Delayed Start-Up

activates by removing the cover and turning the thermostat up and down. You should see the mercury make and break contact as it moves in the bulb.

Wait for up to seven minutes before determining that the unit is not operative. Many units have a timed start-up delay of up to seven minutes.

It is beyond our scope to troubleshoot compressor failures. This information is intended to give you an appreciation for the complexity of the issue. If the compressor doesn’t operate, just record it and recommend further investigation.

IMPLICATIONS
STRATEGY

12.2.8
Inadequate Cooling

Poor comfort and high energy costs are the implications of inadequate cooling.

Measure the temperature drop across the evaporation coil, looking for a 15°F to 20°F drop, roughly. Also, sense how comfortable the house is, if inspecting on a warm day.

12.3
CONDENSER COIL

Function

The outdoor condenser coil transfers heat from the refrigerant into the outdoor air. The refrigerant enters the condenser coil from the compressor as a hot (e.g.,150°F) gas. The gas condenses to a liquid as the 95°F outdoor air (for example) is pushed past the coil by the condenser fan. The air coming off the condenser coil may be 110°F.

Beyond the Standards

The condenser coil is in a cabinet often call the **condenser unit** although it contains the compressor and the condenser fan as well as the condenser coil. This cabinet is not normally opened or serviced by the homeowner. Live 240-volt electrical connections are accessible inside the cabinet. Opening this cabinet is beyond the scope of a home inspection.

Receivers

Removing the access panel allows you to examine the compressor shell and determine the compressor age from its data plate. You can also look for a failed capacitor (bulging or leaking), oil leaks (which indicate refrigerant leaks) or corroded and obstructed coil fins, for example.

Noisy

Condensers are typically copper tubes with aluminum fins. The bottoms of some condensers are liquid receivers that collect the condensed refrigerant. Some receivers are separate from the condenser. Some modern coils are all aluminum; these may have a shorter life span than copper tube coils.

Conditions

The condenser unit can be noisy. Its location should take this into consideration. A condenser unit that is adjacent to a patio for example, may be a nuisance.

Common coil problems include the following:

1. Dirty
2. Damaged or leaking
3. Corrosion
4. Clothes dryer or water heater exhaust too close to condenser

12.3.1 Dirty

The outdoor coil sees unfiltered air pass through it when the system is working. Many modern systems draw air in through the sides and discharge through the top. Other intake and discharge arrangements are used.

IMPLICATIONS

Restriction of airflow through the coil leads to poor heat transfer between the coil and the outdoor air. This means the refrigerant leaving the condenser coil will be warmer than it should be. This means that less heat can be removed from the system. Comfort will suffer and energy costs will be high. Over the long term, compressor damage may result.

STRATEGY

The condenser coil should be inspected at least from the outside of the condenser cabinet. Many inspectors use a light to ensure that there is free air passage through the coils. Where there is dirt on the coils, cleaning should be recommended as regular maintenance.

12.3.2 Damaged/Leaking

It is common to find the fins on condenser coils damaged. Grilles or louvers are often damaged, and occasionally you'll find mechanical damage to the cabinet itself.

IMPLICATION

Damaged fins reduce airflow and heat transfer, adversely affecting system performance, comfort and costs.

STRATEGY

Look at as much of the fin surface as you can. If the coil is clogged, remove some of the dirt to determine if the fins are bent. Straightening the fins is possible, but leave it to a service person.

Oil Stains

Coils may leak if they are damaged. You won't see refrigerant leaking (it's a colorless gas) but you may see oil stains below the condenser coil. This oil is carried with the refrigerant, and does appear on and below coils or pipes where there are leaks.

12.3.3 Corrosion

Corroded fins are common, particularly on older units.

IMPLICATIONS

The implications of a corroded coil are the same as those of a dirty or damaged coil.

STRATEGY

The inspection strategy is the same as looking for a dirty coil or damaged fins. Again, some inspectors use a light to look through the coil.

Make sure the condenser is sitting on a pad so that it is above the soil. A cabinet that is partially buried in dirt will be subject to accelerated corrosion.

Look for corrosion on the cabinet and its grilles or louvers.

12.3.4 Clothes Dryer or Water Heater Exhaust Too Close to Condenser

IMPLICATIONS

Condensers often are clogged with lint from a clothes dryer discharging nearby.

The lint will clog the fins and the warm air from the dryer or water heater makes it hard to dump the heat outdoors. The fins may be corroded by the acidic water heater exhaust.

STRATEGY

Dryer and water heater vents should be about 6 feet from the condenser. Look for lint on the coil. Recommend moving the vent if it's too close to the condenser.

Inspector in the House: Uncovering the Truth

Many homeowners in our area cover the outdoor condenser coil of their air conditioners in the winter. While some specialists say this may lead to condensation and corrosion, that's another discussion. This simple story is to help you avoid doing what we did.

One of our newest inspectors was on the case in May. The weather was in the high 70s, but it was raining. We usually start our inspections outside and then move into the house. But the hard rain made the inspector start the inspection inside, thinking the rain would probably let up.

Testing the air conditioning is a normal part of the process, and the inspector flipped the thermostat control from Heat to Cool and turned on the air-conditioning unit. He did not know there was a cover on the outdoor unit. Not surprisingly, when the outdoor unit cannot move any air across the condenser coil, the unit does not work very well.

The screeching noise from outside alerted the inspector and everyone within earshot of the problem, as the strap that secured the cover in place was tangled in the fan blades. We were lucky that it only cost us a service call, since there was no major damage done.

What did we learn? The obvious thing is to check the outdoor unit before running the cooling system. The other mistake the inspector made was moving the control from Heat to Cool and starting the unit. We should not be the first to operate a cooling system at the beginning of the season. The refrigerant may have leaked out over the winter and the compressor may be ruined by running the unit. The compressor sump heater may not be on, meaning that there may be liquid refrigerant mixed with the lubricating oil. This, too, may ruin the compressor.

If the system has not yet been operated in cooling mode (a good assumption if the thermostat is set to Heat), recommend servicing before the unit is used.

12.4 EVAPORATOR COIL

Function

The evaporator coil transfers heat from the house air into the refrigerant in the coil. The refrigerant boils inside the coil and the house air temperature drops (e.g., from 75°F to 60°F) as it is pushed past the coil by the house air fan.

Copper Tubes

Evaporator coils are usually copper tubes with up to 14 aluminum fins per inch of tube. This type of coil provides good surface area for heat transfer and allows the coils to be compact. Some modern systems have all aluminum coils, which may have a shorter life than copper tube coils.

A-Coils

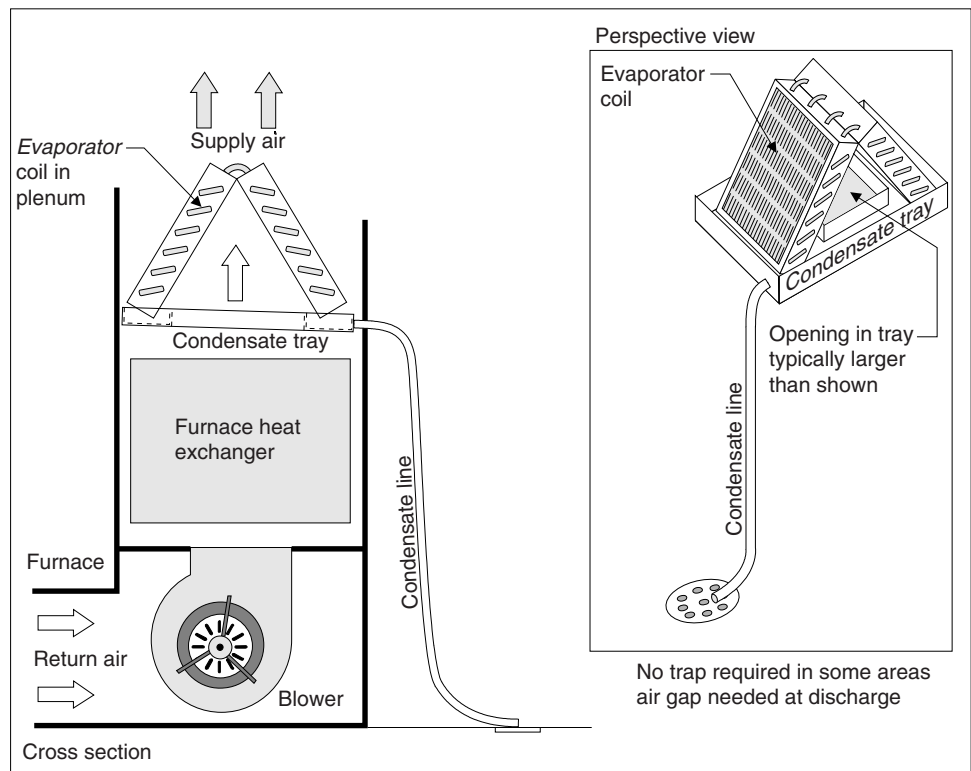
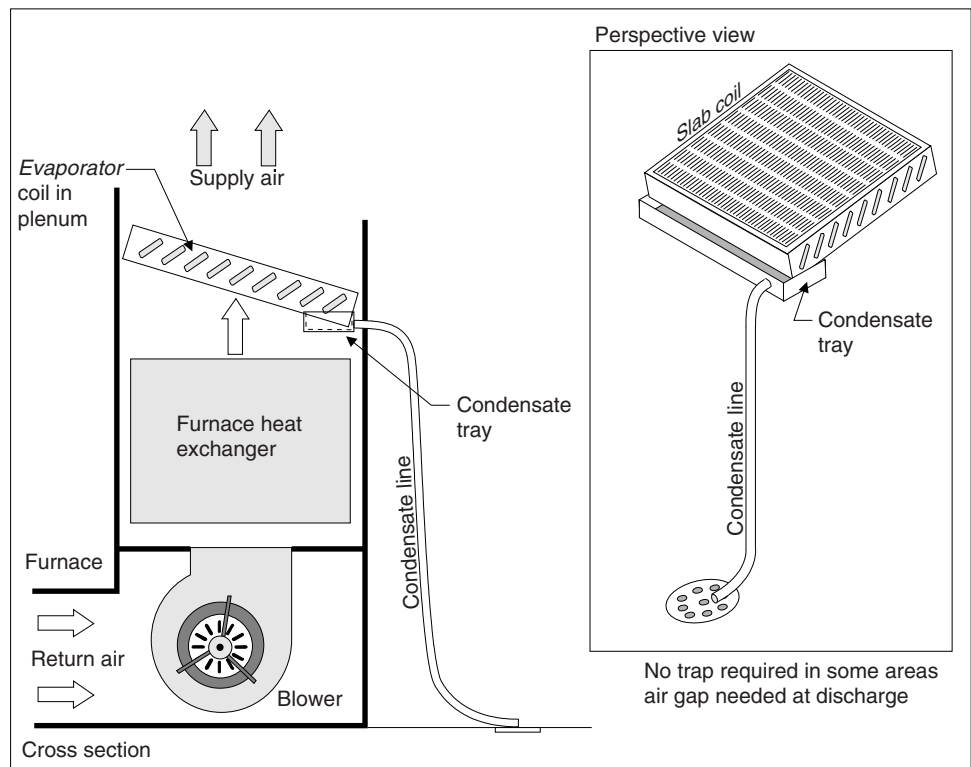
The evaporator coil is often an **A-coil** installed with a condensate tray below the bottom of each leg of the "A" (see Figure 12.6). These trays catch the condensate coming off the coil and allow air movement through the coil. The air passes up between the trays through the coil.

Slab Coils

Slab coils (Figure 12.7) are installed on a small angle with a single condensate tray below the lower end of the coil.

Vertical Coils

If the unit is arranged so that airflow across the coil is horizontal, the coil can be vertical with a tray at the bottom.

FIGURE 12.6 A-Coil**FIGURE 12.7** Slab Coil

Coil Location

Houses with forced-air heating typically have the evaporator coil just past (downstream of) the furnace heat exchanger. The coil may be in a basement, crawl-space, attic or closet. Houses without a forced-air furnace are often also houses without basements. The evaporator coil, fan and much of the duct system are often in the attic.

Downstream of Furnace Heat Exchangers

Evaporator coils must be downstream of (after) gas, oil or propane furnace heat exchangers. Heat exchangers would rust out quickly if the coils were upstream. The coils can be either before or after electric heating elements.

Conditions

Common evaporator coil problems include the following:

1. No access to coil
2. Dirty
3. Frost
4. Top of evaporator dry
5. Corrosion
6. Damage

12.4.1 No Access to Coil

On many installations, there is no access port in the ductwork to inspect and service the coil.

IMPLICATIONS

If the unit can't be accessed for inspection and cleaning, it will get dirty and suffer from a lack of maintenance. It's also possible that condensate overflow or leakage may occur and go undetected for some time. If the coil is above the furnace heat exchanger (common in an upflow furnace), considerable rusting of the heat exchanger can occur from condensate dripping on to the heat exchanger.

STRATEGY

In some areas, the coils are always accessible. In other areas, coils are rarely installed with an access cover. Find out what is the norm in your area, and if the coils are usually accessible, write up any that are not. Whether the coils are typically accessible or not, it's probably safe to say they need cleaning if there is no access.

When coils are in the attic, access is often awkward and unpleasant, especially on a hot day. However, it is part of a Standards inspection to check these.

12.4.2 Dirty

Evaporator coils have fins that are easily clogged.

IMPLICATION

Dirt on the coil restricts airflow through the coil and inhibits heat transfer across the coil, resulting in inadequate cooling and high operating costs.

Another implication is the possibility that the refrigerant returning to the compressor may be too cold and in a liquid state. The compressor may be damaged if it pumps liquid rather than gas.

STRATEGY

Look at the coil to see if it's clean. If possible, look at the upstream side of the coil where the dirt will typically accumulate. Sometimes a mirror and flashlight are necessary to get a good look. The dirt often accumulates in a mat that can be rolled off the coil, much like the lint that accumulates on a clothes dryer filter.

In some cases, mold may also be noted on the coil.

12.4.3 Frost

If frost is seen on the coil, service is necessary.

System performance will deteriorate and compressor damage may result from the suction line containing liquid rather than gas.

Check the coil for frost. Although no frost should be visible, if there is a small amount of frost on the inlet port of the coil, this may not be a serious problem. Servicing should, however, be recommended.

If frost is visible on other parts of the coil, compressor damage is a distinct possibility and the system should be shut down and recommended for service.

IMPLICATION

STRATEGY

12.4.4 Top of Evaporator Dry

When the evaporator is operating, the entire surface should be covered with condensation.

If this is not the case, poor comfort and system stress can result.

If you can look at the coil when the system is or has been operating, check for uniformity of condensation on the coil. This is a secondary indicator to checking the temperature split across the coil. If the split is appropriate, there may not be a problem.

IMPLICATIONS

STRATEGY

12.4.5 Corrosion

The copper or aluminum fins may corrode.

All coils will corrode as they age, and as long as there is not a significant reduction of heat transfer or airflow, this is not a major concern.

Corrosion caused by system contaminants is much more serious. These contaminants are usually left over from an improper evacuation of the refrigerant when a component was replaced. The contamination leads to refrigerant line failure, expansion device blockage and corrosion of the soldered joints.

The coil should be examined through the access port, paying particular attention to the joints at the expansion device and the connection between the refrigerant tubing and the coil. Corrosion of the copper will be green or blue. If the problem is system contamination, the corrosion will be uniform around the joint.

IMPLICATIONS

STRATEGY

12.4.6 Damage

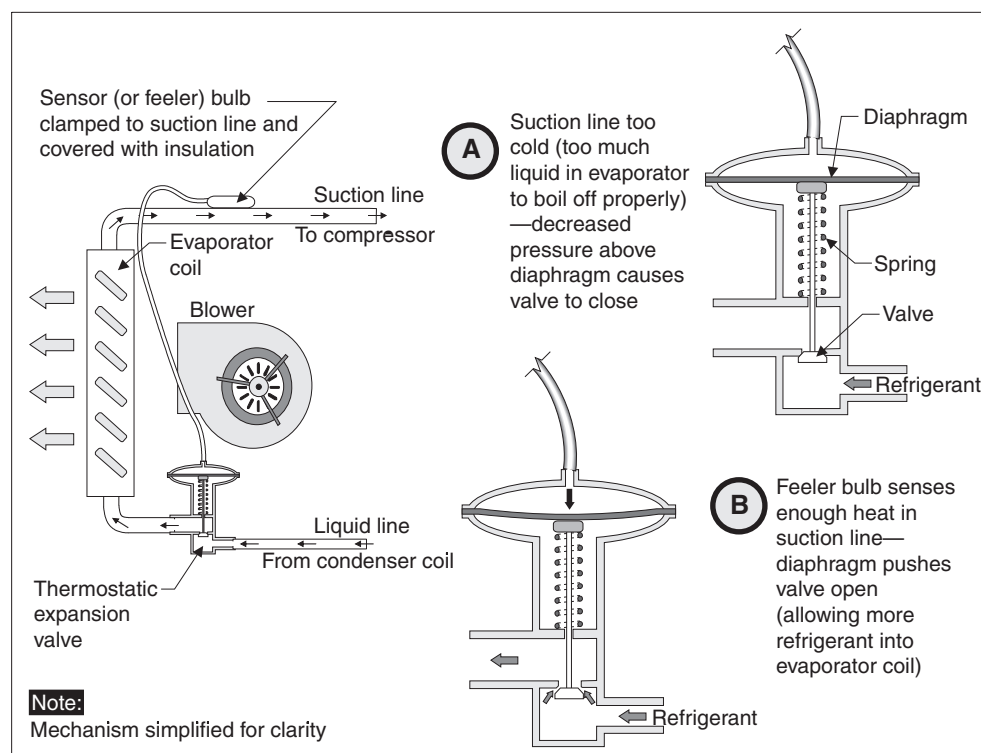
The fine aluminum fins on the coils can be damaged by aggressive cleaning. If the fins are bent, the airflow will be restricted.

Such damage can cause reduced airflow and comfort in the house, reduced heat transfer and possible compressor damage.

When inspecting the coil, look for damage to the fins. In some cases, damaged fins can be straightened.

IMPLICATIONS

STRATEGY

FIGURE 12.8 Thermostatic Expansion Valve

12.5 EXPANSION DEVICE (METERING DEVICE)

Function

This is the device located just upstream of the evaporator coil that changes the refrigerant from a high pressure, high temperature liquid to a low pressure, low temperature liquid.

Capillary Tube

One common expansion device in residential air conditioners is a **capillary tube**. This is a very small-diameter copper tube wrapped into a coil. It simply creates a bottleneck in the liquid line. As the refrigerant comes out of the end of the small-diameter capillary tube, it expands into the larger tube. This lowers the pressure and, consequently, the temperature.

Thermostatic Expansion Valve

Another expansion device is the **thermostatic expansion valve (TXV or TEV)**, as shown in Figure 12.8. The TXV is usually found on air conditioning systems larger than three tons, or on combination heat pump and electric furnace packages. The TXV is a more precise metering device than the capillary tube and allows the system to adjust to its environment so that the compressor should never receive liquid refrigerant.

Location

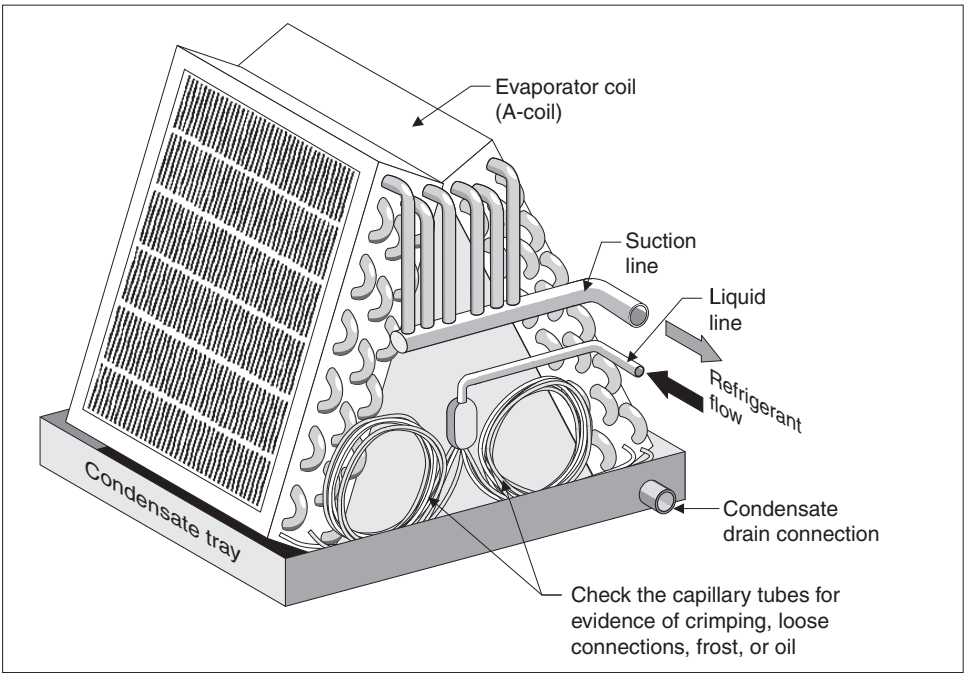
The expansion valve is often visible in the refrigerant lines just outside the duct.

Conditions

Common expansion device problems include the following:

1. Capillary tube defects
2. Thermostatic expansion valve connections loose
3. Clogged orifice
4. Expansion valve sticking

FIGURE 12.9 Capillary Tube Defects



12.5.1 Capillary Tube Defects

The capillary tube may be crimped, disconnected, frosted, or covered with oil (Figure 12.9).

IMPLICATION
STRATEGY

Poor cooling performance will result from capillary tube defects.

If visible through the access port in the duct system, look at the tube for evidence of crimping, loose connections, frost or oil.

12.5.2 Thermostatic Expansion Valve Connections Loose

The TXV may have a loose bulb or loose liquid line connection. The liquid line connection point may also be cracked.

IMPLICATION
STRATEGY

These conditions will result in loss of refrigerant and poor system performance.

Look for evidence of a loose bulb or liquid line connection. Look for evidence of cracking at the liquid line connection.

12.5.3 Clogged Orifice

IMPLICATIONS
STRATEGY

If the expansion device is plugged, the refrigerant can't flow and the system will shut down, ideally before the compressor is damaged.

Troubleshooting air conditioning systems is beyond our scope, but frost near the expansion device may suggest an obstruction.

12.5.4 Thermostatic Expansion Valve Sticking

IMPLICATIONS

The valve components may not move properly, resulting in no refrigerant flow.

STRATEGY

Again, this may not be identified as the reason for poor system performance or non-performance, but this is beyond our scope.

12.6 CONDENSATE SYSTEM

There are several components to the condensate system.

12.6.1 Condensate Drain Pan (Tray)

Location and Function

The **condensate drain pan**, located below the evaporator coil, catches the condensate that drips off the outside of the coil as the warm moist house air passes over the coil. The cold refrigerant in the coil cools the air, causing the moisture to condense on the outside of the coil. This is how air conditioners dehumidify the house air.

If there is access to the coil, the condensate drain pan can be inspected. If there is no access, you won't be able to see the pan, and as mentioned earlier, you should recommend servicing and the provision of an access panel so that the coil and pan can be inspected and cleaned. The pan should be slightly sloped so water flows to the drain connection. The condensate is piped to a floor drain, sink, outdoors or another suitable discharge point.

Conditions

The common drain pan problem is leaking.

IMPLICATIONS

Leaking pans allow water damage to occur to whatever is below. In many areas, the evaporator coil is directly above the furnace heat exchanger. A leaking pan can rust out the furnace heat exchanger, rendering the furnace ineffective and creating an unsafe situation, with a risk of carbon monoxide entering the house.

STRATEGY

Look at the pan for evidence of leaking, including cracks, rust, dirt accumulation, a drain line obstruction, improper slope, or poor attachment. Look for staining on the outside of the pan and on whatever components are below.

If there is no access cover, look around and below the cabinetry and furnace components below the coil for evidence of water streaking, staining or rusting.

If the air conditioner is running or has been running recently, look for moisture.

12.6.2 Auxiliary Condensate Drain Pan

Second Line of Defense

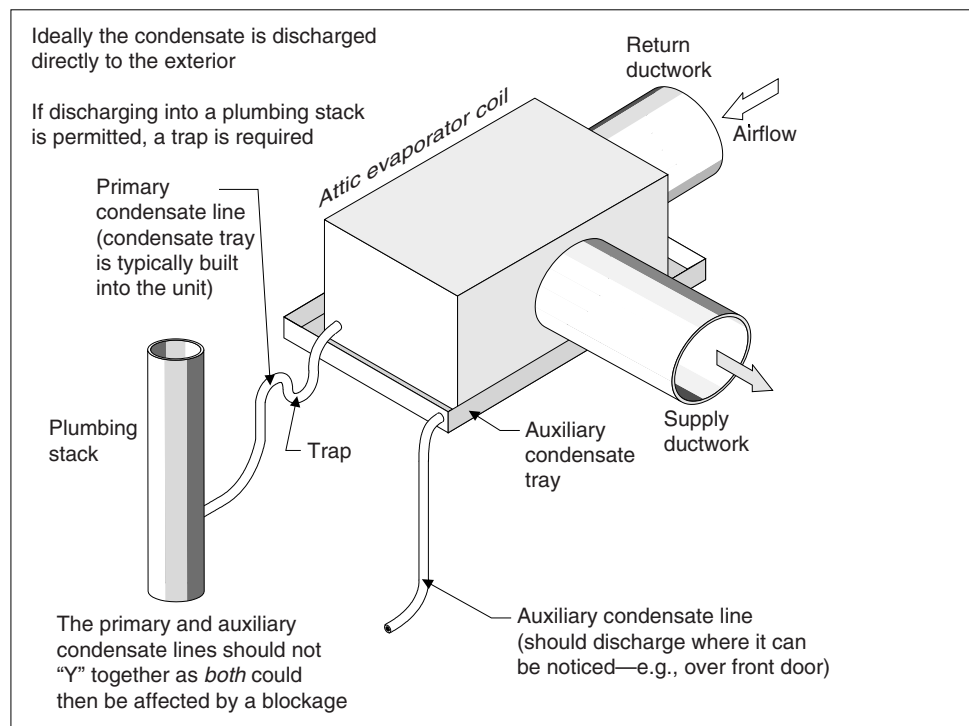
Auxiliary pans are typically provided where an evaporator coil is located in an attic or anywhere above finished living space (see Figure 12.10). These are only used when failure of the main drain pan will damage the house. This is a secondary pan, provided because the primary drain will eventually leak.

Causes of Leaks

This pan should be checked for the same conditions as the main condensate drain pan, including rust, dirt accumulation, a blocked drain line, improper slope and poor attachment.

Float Switch

Some drain pans have a float switch that will shut off the system if the water level rises too high, instead of an auxiliary pan. This switch may shut off just the compressor or the compressor and the blower. This switch can be tested by hold-

FIGURE 12.10 Auxiliary Condensate Line from Attic Evaporator Coil

ing it up for roughly five minutes to see whether the unit shuts off. In some cases the whole system shuts down. In some cases only the compressor and condenser fan shut down.

12.6.3 Condensate Drain Line

Function and Material

This drain line carries condensate from the pan to an appropriate discharge point. The line is typically plastic or copper and is usually $\frac{3}{4}$ -inch diameter. In some areas, the condensate line is insulated to prevent it from sweating as it runs from the evaporator coil to the discharge point.

Separate Drain Pipe Needed for Auxiliary Pan

An auxiliary drain should have a separate drain line from the pan that is not manifolded with the line from the primary pan. A clog in one condensate drain line should not render both pans ineffective. There should be no trap in the auxiliary drain line, which is typically $\frac{3}{4}$ -inch diameter.

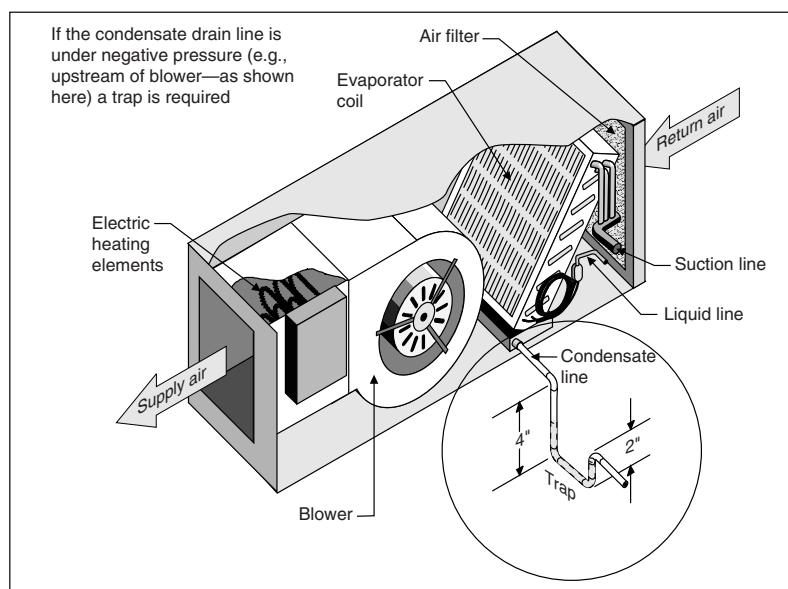
Discharge Point

The discharge point from a primary pan should be out of the way. The discharge point for a secondary pan should be in the way, and hard to ignore; over the front door is good. It should catch your attention, because something is wrong if it's working.

Conditions

Common drain line problems include these:

1. Leaking
2. Blocked
3. No trap
4. Improper discharge point

FIGURE 12.11 Trap Required in Condensate Line**IMPLICATIONS****STRATEGY****Leaking**

A leak will allow condensate to flow where it was not intended. The implications are similar to those we discussed with respect to leakage from the condensate drain pan.

Inspect the length of the condensate drain line looking for active leakage if the unit is running or has been running recently. If the air conditioner is idle, look for evidence of staining, streaking, loose connections or split piping.

Blocked

The implications of blockage are similar to those caused by a leak in the drain pan.

If the unit is operating, make sure water is discharging out of the drain line. If the unit is at rest, pouring water into the drain pan and checking the discharge point will reveal blockages, although this goes beyond the Standards. In some cases, there is a vent (relief opening) downstream of the trap in the condensate drain line. Watch that this is not discharging. If water is coming out of this relief opening, there is probably a blockage in the trap.

Blockages are common at the entry and exit points at the drain line.

No Trap

Traps are often required on condensate drains, especially when the condensate pan is upstream of (before) the house air fan (Figure 12.11). This is typical of air-conditioning-only systems, or air-conditioning-plus-electric-heat. The water in the pan is under negative pressure because it is on the suction side of the pan.

If the line is under negative pressure, air may be sucked up the drain and the water may not be able to flow down the drain line. In some cases, the drain pan may fill up and overflow. A trap prevents this. Standard traps have a 4-inch drop on the upstream side, and a 2-inch rise on the downstream side. Traps in condensate lines are required in many jurisdictions. In some jurisdictions, if the condensate line is permitted to drain into a waste plumbing pipe, a trap will be required. Check with authorities in your area to determine whether air conditioning condensate lines

Purpose—Negative Pressure

Not into Plumbing Stack

IMPLICATIONS

STRATEGY

Discharge Outside

Sink, Floor Drain, Plumbing Fixture

Through Floor

Into Waste Plumbing Stack

IMPLICATIONS

STRATEGY

Conditions

IMPLICATION

require traps and, if so, whether there are any specific rules about where and how they are installed. Traps are usually outside the evaporator plenum, as close to the drain pan as possible.

If the condensate line does terminate in a waste plumbing stack, the trap will prevent sewer odor and bacteria from entering the house air supply, at least until the trap water evaporates. However, this discharge point is not allowed in most areas.

Some systems without traps work most of the time. During hot humid weather, the drain pan may overflow. Most would agree that if a condensate line discharges directly into a waste plumbing system, a trap is necessary for health reasons. Incidentally, when the air conditioner hasn't been run for several months (over the winter), the water in the trap will evaporate, and for this reason, direct connection to a waste plumbing stack is not a good idea even if permitted in your jurisdiction.

Trace the condensate line from the drain pan to its discharge point. It should be obvious whether or not a trap has been provided.

Improper Discharge Point

The best place for air conditioning discharge is usually considered to be outside the building onto the ground. The discharge should not run down the exterior of the wall.

Some jurisdictions allow the condensate to drain into a sink or floor drain as long as there is an air gap to prevent siphoning. It may be permitted to discharge the condensate into the tail piece of a basin (upstream of the trap) or into the overflow for a bathtub (again, upstream of the trap). Check with authorities to find what is allowed in your area.

In some areas, the condensate drain line goes down through the floor slab and discharges into the granular material below. This is not considered a good arrangement because you can't see the condensate line working. An obstruction may go undetected until considerable damage is done. Sub-slab gases (such as radon) may also find their way into house when the unit is not running.

We also recommend against discharging the condensate drain line into the waste plumbing stack. There is the issue of a blockage going unnoticed for some time and there is also the possibility of sewer odor backing up into the house air system, as discussed earlier.

Non-performance that goes unnoticed may cause water damage. If discharging into waste plumbing, sewer odor and its associated health issues are other potential implications.

Trace the condensate discharge line to its termination point. Ensure that the end of the line is open, and if the air conditioner is operating, water should be coming out.

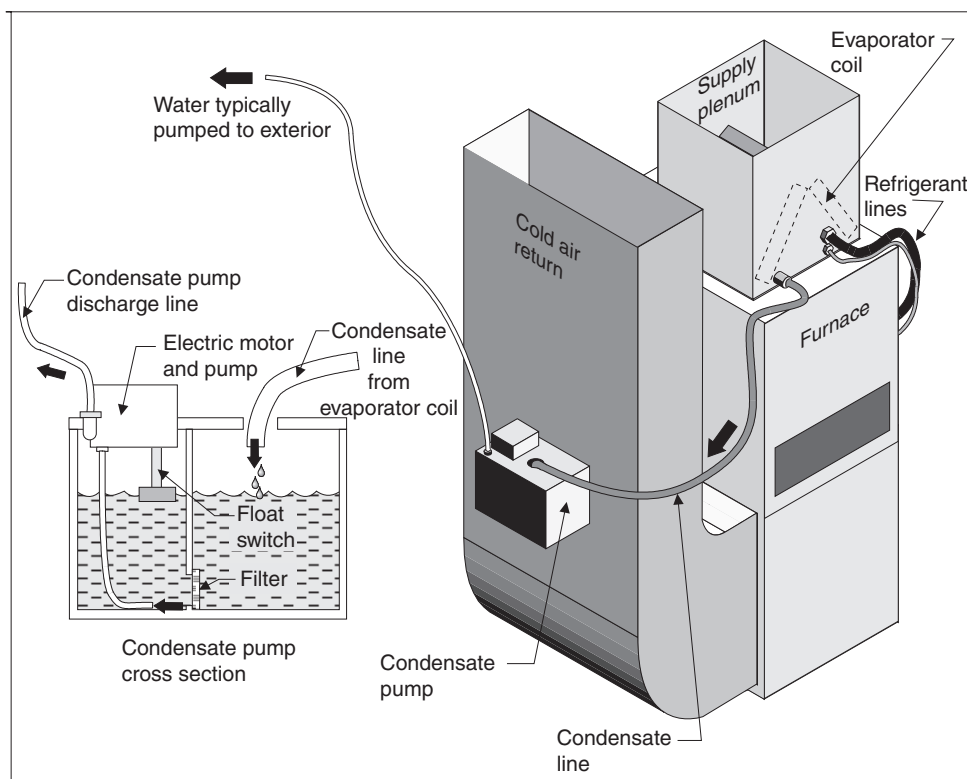
12.6.4 Condensate Pump

If the condensate cannot run from the evaporator coil outside by gravity, and if there is no drain or sink nearby that the condensate can be discharged into, a pump is required (Figure 12.12). Condensate pumps are typically low-quality sponge pumps.

Pumps may be inoperative, leaking, or poorly wired.

Inoperative

Since the pump is located outside the evaporator coil housing and duct system, the implication of pump failure is water spilling onto the area around the pump and reservoir.

FIGURE 12.12 Condensate Pump**STRATEGY**

If the system is operating, you can look for water discharging from the condensate line and listen for the pump working.

If the pump does not work, the pump may be inoperative, although you have to be careful. The electrical power to the pump may have been shut off when the air conditioner was shut off for the season. If you're looking at an air conditioning system outside of the cooling season, the condensate pump may not be powered. Be careful not to describe the pump as inoperative under these circumstances.

IMPLICATIONS**Leaking**

Leakage may result in water damage and a burned-out reservoir pump.

STRATEGY

Check the pump and the area around it.

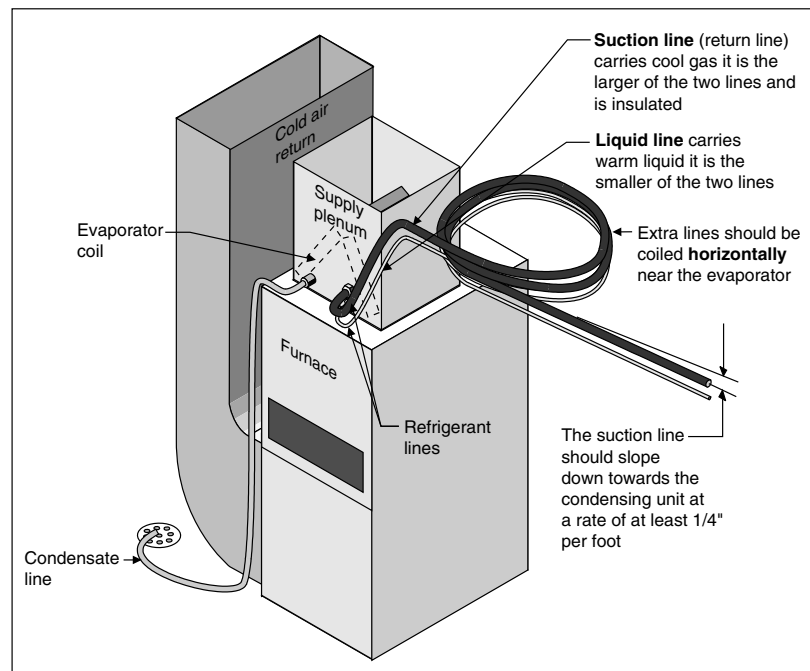
Poor Wiring

Many potential electrical problems can be found at the condensate pump. Because of the proximity of the water and electricity, condensate pumps should be grounded. Many are not. Other common electrical conditions include loose connections, poorly supported wiring, open connection boxes, and overfusing.

Poor wiring can result in fire or electrical shock.

IMPLICATIONS**STRATEGY**

Check the electrical supply to the condensate pump as you check the rest of the electrical system. Ensure that the pump is grounded.

FIGURE 12.13 Refrigerant Lines

12.7 REFRIGERANT LINES

Function and Material

The lines that carry the refrigerant between the evaporator and condenser coils and through the compressor and expansion device are typically copper.

Suction Line

The larger line typically carries a cool gas and is insulated. This is referred to as the **suction line**. It is also called the **return line**.

Liquid Line

The smaller uninsulated line typically carries a warm liquid. It is most often called the **liquid line**.

Allow Oil to Flow Back to Condenser

Where the evaporator coil is higher than the condensing unit, the suction line should slope down toward the condensing unit with a slope of at least one quarter inch per foot.

Line Coiled Horizontally

Extra lines are usually coiled near the evaporator coil. The line should be coiled horizontally rather than vertically (Figure 12.13). The coils should allow oil to flow down through the coil and back to the condensing unit.

Pressures in the lines may range from 50 psi up to 275 psi. Since the refrigerant is a gas at atmospheric temperature and pressure, if it leaks it will dissipate as a gas. It may leave an oil residue.

Filter/Dryer

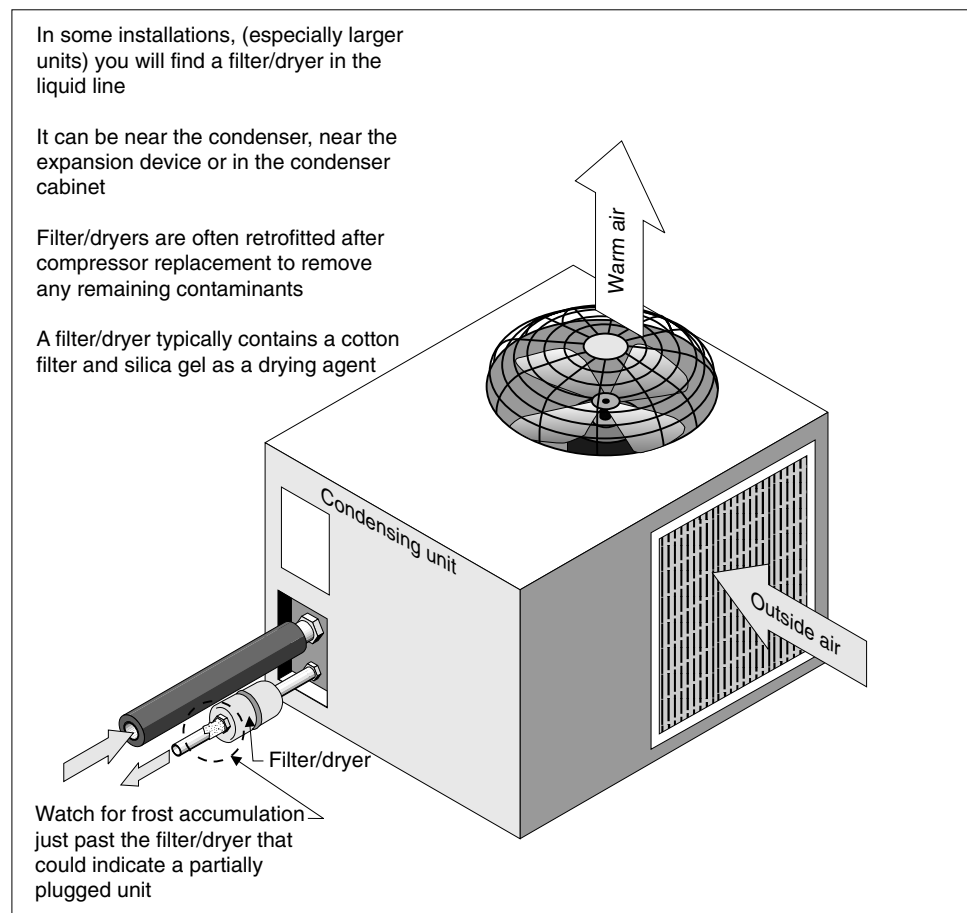
In some installations, you will find a **filter/dryer** in the liquid line (Figure 12.14). Filter/dryers clean and dry the refrigerant. They are often added to a system where the compressor has been replaced. They help remove any contaminants. They are roughly the size and shape of a soft drink can. They may be located in the liquid line near the condenser outlet or near the expansion device.

Watch for Frost

Frost accumulation just past the filter/dryer indicates a partially plugged unit and service should be recommended.

Supports and Radius of Bends

Supports for refrigerant lines should be every 5 to 6 feet. Bends in refrigerant lines should have a minimum 12-inch radius.

FIGURE 12.14 Filter/Dryer on the Refrigerant Line*Conditions*

Common refrigerant line problems include the following:

1. Leaking
2. Damage
3. Missing insulation
4. Lines too warm or cold
5. Lines touching each other

12.7.1 Leaking

A leak in a line is usually identified by the oil residue on the line. Because oil travels through the system with the refrigerant, a leak will often show up as an oil stain. Escaping refrigerant boils off and leaves no trace, other than the oil that is left behind.

IMPLICATIONS

If the refrigerant leaks out, the system performance will deteriorate and comfort levels in the house will decrease. Ultimately, the system will shut down or the compressor may fail.

STRATEGY

Since refrigerant line joints are usually only at the coils, concentrate on the connections to the coils at either end, where visible. If there is a thermostatic expansion valve, check its connections.

Check along the length of the line for evidence of mechanical damage, particularly in exposed areas and at the interior and exterior penetration points through the house wall. Refrigerant lines need support so that joints aren't stressed under the weight of the lines.

Lines through the Wall

On attic units, pay attention to where the lines disappear into walls or ceilings.

Where refrigerant lines go through walls, the hole in the wall should be considerably larger than the refrigerant lines and the lines should be in conduit where they go through the wall. The ends of the conduit should be sealed with a flexible material at either end to allow movement but to prevent moisture and insect entry into the building and heat loss out of the building.

Oil Stains

Look for evidence of oil stains on the refrigerant lines.

12.7.2 Damage

In some cases, the lines may be crimped or damaged without leakage. Any reduction in the diameter of the lines will act as a pressure restriction, changing the temperature of the refrigerant that passes through the restricted area. This will adversely affect system performance.

IMPLICATIONS

Such damage results in reduced performance of the system.

STRATEGY

Follow the refrigerant lines, looking for evidence of crimping or flattening, particularly at changes in direction or where the lines go through walls or ceilings.

12.7.3 Missing Insulation

The large-diameter line (suction line or gas line) should be insulated along its entire length. The insulation performs two functions.

1. It prevents the suction line from sweating and dripping water inside the house.
2. The insulation also prevents the suction line attracting heat from the outdoors on its way to the condenser coil. We are trying to take heat out of the house and dump it outside. We don't want to gather outside heat and dump it into the suction line before it goes into the compressor.

IMPLICATIONS

System performance will be adversely affected if the outdoor insulation is missing. If the indoor insulation is missing, water damage to the home may result.

STRATEGY

Look for the insulation to be intact along its entire length. Pay particular attention to the outdoor section of the suction line.

12.7.4 Lines Too Warm or Too Cold

Cold Suction Line

When the system is operating, the large insulated suction line should be cold to the touch and sweating at any point where there is no insulation.

Warm Liquid Line

The smaller uninsulated liquid line should be warm to the touch after the system has been operating for 10 or 15 minutes.

*Frost***IMPLICATIONS****STRATEGY**

Frost on the suction line may indicate too much refrigerant going through the expansion device, an inoperative house air fan, too much refrigerant or too low an outdoor temperature. Frost on the liquid line may mean the dryer is clogged. Frost on the expansion device may mean the device is clogged with ice.

Lines that are too warm or too cold can indicate a lack of adequate cooling in the house and possible compressor damage.

Touching the lines is a part of any inspection when the air conditioner is operating. Look for the suction line to be at roughly 45°F to 55°F, and the liquid line to be at about 90°F to 110°F. Measurement with instruments is not needed.

If the suction line isn't cold, or the liquid line isn't warm, servicing should be recommended.

12.7.5 Lines Touching Each Other

IMPLICATIONS**STRATEGY**

The liquid and suction lines should not be in contact. If they are, heat transfer between the lines and reduced system efficiency may result.

Make sure these lines don't touch each other.

Inspector in the House: Detective Inspector

I was performing a pre-listing inspection for a seller in the fall. We do a lot of these before the home goes on the market. It was too cold to run the air conditioner but we always do a visual inspection anyway. There was a lot of storage in the furnace room and a pile of firewood outside the house. I had trouble tracing the refrigerant lines from the condenser coil to the evaporator.

Since the client was the seller, I asked if the air conditioner worked. With what may have been the slightest hesitation, she told me the air conditioner had worked fine all summer. As I made my way around the outside, I saw signs of recent excavation along the foundation wall. The homeowner advised they had just had the basement waterproofed to ensure it did not leak. She showed me the listing sheet, which indeed described the recently waterproofed basement.

Moving some firewood allowed me to confirm that the air conditioning lines had been cut for the basement waterproofing to be done! I have not concluded whether or not the homeowner knew about this. Thank goodness we don't have to figure these things out.

By the way, moving the firewood is not required as part of a home inspection, but when you are chasing a clue you sometimes get fanatical about finding the answer.

12.8 CONDENSER FAN

Function

This fan blows outdoor air across the hot condenser coil, cooling the hot refrigerant gas. The gas inside the coil condenses as it cools and the air passing over the coil heats up. This is how we dump the house heat into the outdoor air.

Location and Orientation

The outdoor fan is located in the condenser cabinet. The fan draws air in through the sides of the condenser coil and discharges the warmer air through the top of the cabinet.

Conditions

Common fan problems include the following:

1. Excess noise/vibration
2. Inoperative
3. Corrosion or mechanical damage
4. Obstructed airflow

12.8.1 Excess Noise/Vibration

IMPLICATIONS

Bearing noise is an indicator of a bearing about to fail, usually from a lack of lubrication.

Unbalanced fan blades will rotate but will eventually cause bearing failure. Unbalanced fan blades are usually a result of mechanical damage or dirt and will not move as much air as they should.

STRATEGY

In any case, system failure is the ultimate result of a noisy fan.

Listen for high pitched intermittent or steady squeals from the fan area. Observe the fan blades for vibration while rotating and listen for a helicopter-type noise from unbalanced fan blades.

When the fan is not operating, look at the blades to see if they are caked with dirt or are obviously damaged or misaligned.

12.8.2 Inoperative

IMPLICATIONS

When the outdoor fan doesn't work, very little heat will be transferred to the outdoor air, which means that the air conditioning will not perform properly. Over time, compressor failure may result.

STRATEGY

When the unit is running, ensure the fan blades are turning. Most modern fans are axial and turn at approximately 1,800 RPM. Newer quiet low rpm fans may turn at 875 rpm. If the fan is not moving or is moving very slowly, service is required.

12.8.3 Corrosion or Mechanical Damage

IMPLICATIONS

Rusted or damaged fans may not operate. The implications are the same as for an inoperative fan.

In other cases, the fan may turn but with reduced efficiency or speed. This will move less air across the coil and reduce the heat transfer to the exterior air. This, of course, will also reduce the comfort in the house and make the whole system work harder.

STRATEGY

Look closely at the fan and motor for evidence of rusting. A flashlight may be necessary here. Look also for evidence of mechanical damage to the motor casing and fan blades. Check for cracks in the fan blades.

The wisdom of covering condenser units during the winter months is not universally agreed on. Some say they should be covered; others say they shouldn't. People do agree that if a cover is used, it should be breathable, so it won't trap condensation in the unit.

12.8.4 Obstructed Airflow

Intake air should be unobstructed for 1 to 3 feet adjacent to the unit (depending on the manufacturer's recommendations). The clearance on the exit side or top should be 4 to 6 feet.

The airflow through the outdoor fan may be blocked on the intake or exhaust side. A clogged, damaged or corroded condenser coil can block the air supply to the fan.

Partially or totally restricted fans will lead to non-performance and possible compressor failure.

Look at the airflow into and out of the fan.

Some people maintain that air conditioning condensers should be kept out of direct sunlight. Many manufacturers do not consider this a significant issue. Incidentally, manufacturers of heat pumps consider it an asset if the unit is located in the sun.

When checking the operation of the outdoor fan, the air coming off the fan should be warmer than the ambient air temperature, even on a hot summer day. This is one indication that the system is operating properly.

IMPLICATIONS

STRATEGY

Sunlight or Shade

Discharge Air Is Hot

12.9 EVAPORATOR FAN

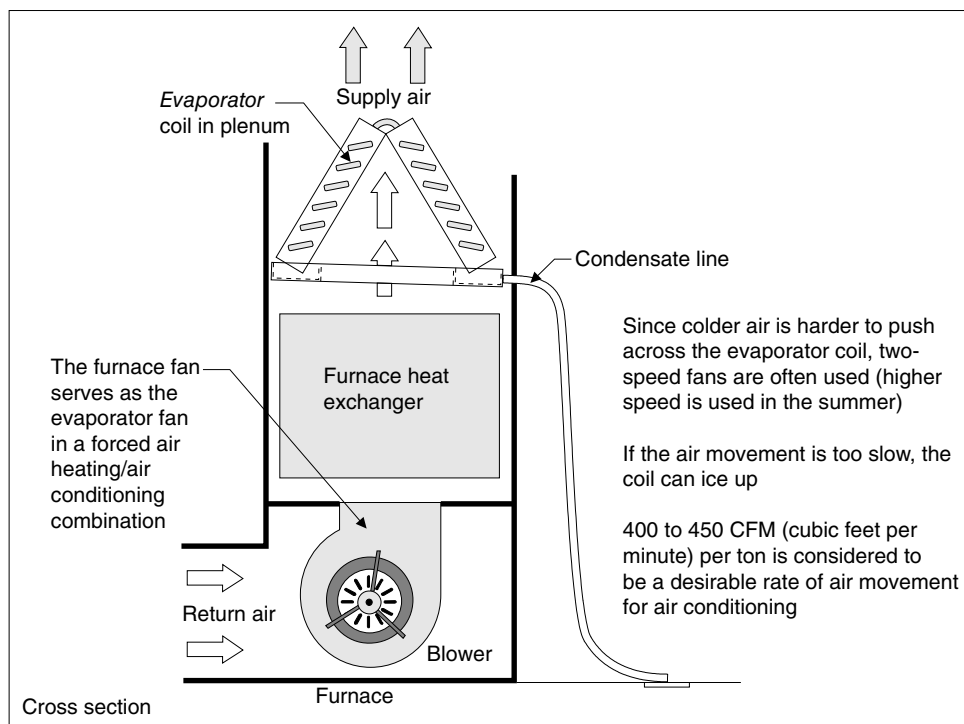
Function

The indoor fan blows house air across the evaporator coil, cooling and dehumidifying the house air (see Figure 12.15).

Too Much Air

The size of the air handling system has to match the cooling capacity of the air conditioning equipment. If too much air is blown across a coil (the air moves by the coil too quickly), there may not be enough cooling to dehumidify the air. Noise

FIGURE 12.15 Evaporator Fan



problems will also develop and moisture will be blown off the coil, rusting ducts and damaging finishes downstream. A maximum air velocity of 500 linear feet per minute is typically recommended.

Too Little Air

If the air moves too slowly, the air may be cooled dramatically, but the air may not move quickly enough through the house to achieve good cooling. Another possible impact of inadequate airflow is icing up of the evaporator coil because the coil sees a colder environment than it should.

Need Twice as Much Air for Cooling

Too little air movement is a more common problem than too much air movement is. Roughly twice as much air movement is needed for cooling as heating with a conventional gas furnace, for example.

Fan Coils Hung from Rafters

Fans and coils in attics are often hung from rafters rather than set on ceiling joists to minimize vibration and noise transferred to the living spaces.

Noisy air handlers or air handlers with a great deal of vibration are usually nearing the end of their life. In some cases, lubrication can restore bearing operation and eliminate noise.

Conditions

Common fan problems include the following:

1. Undersized blower or motor
2. Misadjustment of belt or pulley
3. Excess noise/vibration
4. Dirty fan
5. Dirty or missing filter
6. Inoperative
7. Corrosion/damage

Evaporator fans are the same as those found in furnaces. In fact, when an air conditioner is added to a furnace system, the same fan is used for both heating and cooling. Rather than repeat our earlier discussion, please refer to Chapter 8, *Gas Furnaces*, for more information on fans and their associated problems.

12.10 DUCT SYSTEM

Function

The duct system moves cooled dry air through the house and brings warm moist air back to the evaporator coil for cooling and drying (Figures 12.16 and 12.17). A good duct system keeps all parts of the home at roughly the same temperature.

An air conditioning system relies as much on a good duct system as it does on good cooling equipment. Many air-conditioning systems dramatically under-perform as a result of duct problems.

All of Home Served?

Be very careful to ensure that the duct system feeds all parts of the home. Additions and renovations are sometimes not air-conditioned. Watch for electric heat in a room. This may mean no ductwork and, consequently, no cooling. It is very expensive to add air conditioning to spaces that don't have ducts. You don't want to have to provide a client with air conditioning in rooms because you mistakenly told them that the entire house was air-conditioned.

Watch for houses with hot water heat that have had independent air conditioning systems added. These systems often don't cover the whole house.

High Level Return Grilles

Retrofitted air conditioning systems have different needs than heating systems. Heating systems don't usually require a great deal of return air from the upper floors. Air conditioning systems do need good air return from upper levels.

FIGURE 12.16 Flow of Cooled Air—Older-Style Ductwork

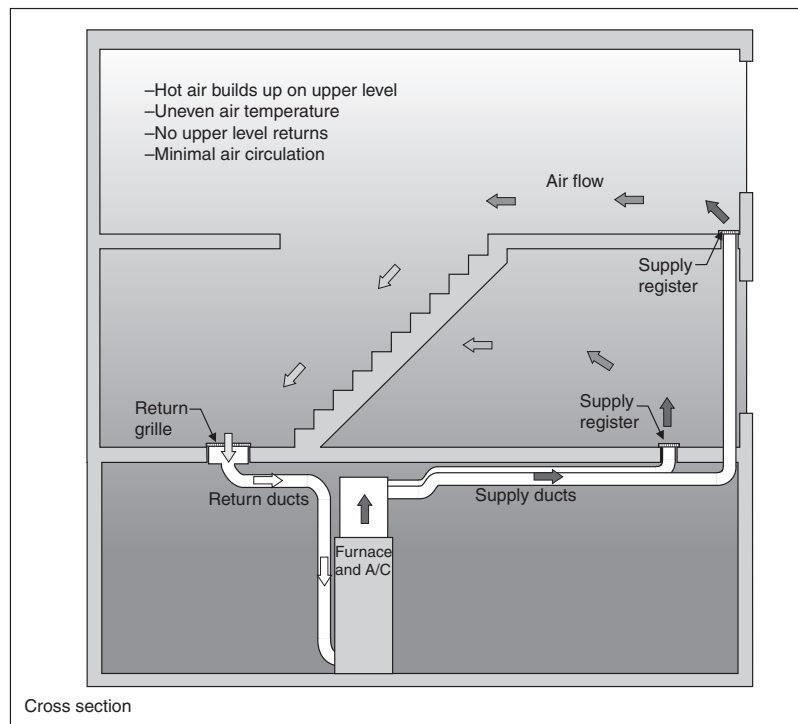
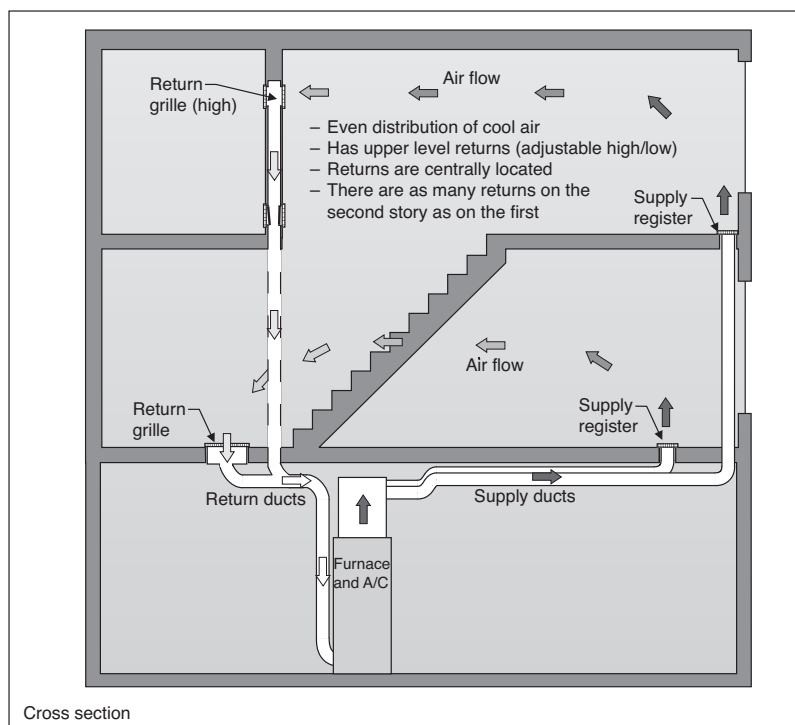


FIGURE 12.17 Flow of Cooled Air—Modern Ductwork



It is common to find large temperature differences between the first and second floors on air conditioning systems that have been added to ducts designed for heating only.

12.10.1 Duct Insulation

Insulation

Ducts passing through unconditioned spaces, such as attics or crawlspaces, should be insulated and airtight. We don't want to collect heat from these areas, or distribute cool air into these areas. As a general rule, the more ducts that pass through unconditioned spaces, the greater is the challenge for the air conditioning system.

Vapor Barriers

The insulation should have a vapor barrier and, like all vapor barriers, it should be on the warm side of the insulation. In an air conditioning environment, this means the vapor barrier should be on the outside of the insulation. The warm, moist air in the attic or crawlspace may condense if it is cooled. We want to keep the attic and crawlspace air warm, but if we inadvertently cool this air, we don't want condensation to form on the outside of the ducts. The vapor barrier is designed to prevent the warm, moist air from getting close to the ducts where it would inevitably form condensation. It also prevents moisture moving through the insulation by vapor diffusion.

The downside to all of this is that putting the vapor barrier on the outside of ductwork that runs through unfinished spaces creates a vulnerable situation. Vapor barriers are often torn.

12.10.2 Supply and Return Registers

Registers are associated with duct problems, but can be looked at slightly differently. A good-sized air conditioning system and reasonable-sized ducts may still result in poor performance if registers are not of the appropriate size, number and location. Conditioned air is heavier than the room air and may be expected to fall. The warmest air is at the top of the room.

An Ideal Arrangement

Introducing conditioned air near the ceiling on an outside wall allows it to fall through the warm air, creating a relatively even temperature throughout the room. The return registers are ideally located on the opposite (interior) side of the room to enhance circulation through the entire room. The return registers should also be high to carry the warmest air out of the room back to the air conditioner to be cooled.

Many systems are not set up this way and, when the same ducts are used for heating and for cooling, compromises are usually made. Heating systems are best set up with the supply and return registers near floor level, just the opposite of what we talked about for air conditioning registers.

Supply on Outside Walls

During the summer, the warmest part of the room is likely to be the outside wall, especially if there are windows in the wall. It makes sense to introduce the coolest air close to the outside wall. Therefore, the supply register should be located near the outside wall.

Returns on Interior Walls

The return registers should be on an opposite and interior wall. In this regard, heating and air conditioning systems have the same register location goals. For the heating system as well, the supply register should be located near an outside wall (usually below windows) and the return register should be located on an opposite and interior wall.

Heating is the Priority

Real-life compromises often mean that registers are set up with heating as the priority over air conditioning. For example, in a two-story house with the ducts designed primarily for heating needs, the return air grilles typically collect cold air that falls down stairwells to the lower parts of the home. Consequently, the return ducts and grilles are usually located centrally, near the bottom of stairwells, and there is often more air return on the lower level than the upper level.

An ideal return air system for air conditioning would have more return air ducts and grilles on the upper level, and they would be located near the top of the second floor. These grilles would be centrally located and above stairwells.

Cooling Better on First Floor than Second

Since most systems are set up with heating as a priority, it is very common to find houses where the air conditioning system works reasonably well on the first floor, but not nearly as well on the second floor.

Conditions

Common duct problems include the following:

1. Undersized or incomplete
2. Supply and return register problems
3. Dirty
4. Disconnected or leaking
5. Obstructed or collapsed
6. Poor support
7. Poor balance
8. Humidifier damper missing

Further discussion of these problems can be found in section 8.9.5 of Chapter 8, *Gas Furnaces*.

Inspector in the House: Appearances Can Be Deceiving

The house is centrally air conditioned. That's what we assume when we see an air conditioning system connected above the furnace. Sometimes we assume too much.

We assumed too much in the house with the garage that had been converted to a family room. It is difficult to add ductwork to an existing slab-on-grade structure. Adding heating with electric baseboards is relatively easy. Adding air conditioning is tougher. We had to pay for an independent air-conditioning system for this room, because we reported the house as air conditioned and failed to mention there was no air-conditioning for the family room.

We also assumed too much in the 1920s house that was open concept on the first floor. In our area, these homes originally had warm air heating with supply ducts running up the interior walls. When the first floor walls are removed, so are the ducts. The second floor registers often remain and electric heaters are used to keep the second floor comfortable in the winter. But there is no air conditioning, and no easy way to add it.

The worst miss I can remember is the beautifully finished basement apartment with supply registers in the ceramic tile-covered concrete floor. The problem was evident if you lifted the register and looked down. There was no ductwork—only dirt. It's one thing to learn after the inspection that the ducts buried in concrete are rusted, flooded, or collapsed. It's quite another to realize there are no ducts at all. This miss cost us \$2,000.

12.11 THERMOSTATS

Conditions

We discussed thermostats in the context of furnaces and won't repeat the information here. You may want to review Section 8.7.

Common thermostat problems include the following:

1. Inoperative
2. Poor location
3. Not level
4. Loose
5. Dirty
6. Damaged
7. Poor adjustment/calibration

Inoperative

Here, we will discuss only inoperative thermostats. The other conditions, and strategies for identifying them, are covered in Chapter 8.

To activate an air conditioning system, the thermostat is turned to a lower temperature than the room temperature. If the air conditioner does not respond after about seven minutes, the system should be described as inoperative.

The thermostat is one component that may cause the system not to operate. There are many other problems that may result in the failure to respond to the thermostat. It's beyond the scope of a home inspection to troubleshoot air conditioning systems, although a bad thermostat can be included as one of the possibilities.

IMPLICATIONS

The air conditioning system won't respond if the thermostat isn't working. It may not start or it may not shut off.

STRATEGY

Turn the thermostat down to see if the system operates. If it does not, you may remove the cover (beyond the Standards) and check that the mercury bulb switch moves properly and that the contacts are working. You may also verify that the low voltage wiring to the thermostat is properly connected. Beyond this, there isn't much you can do.

Calibration

It's possible that the thermostat works, but the calibration is wrong. You may not know this during the inspection, and an analysis of the thermostat calibration is beyond the scope of your inspection.

Indoor Fan Starts Immediately

The air conditioner controls are a little different from the furnace control. When the furnace is activated, the burner typically comes on immediately, but the fan will not come on until the furnace is up to temperature. With air conditioning systems, the indoor fan comes on as soon as the compressor and outdoor fan starts.

12.12 LIFE EXPECTANCY OF COMPRESSORS

Although predicting life expectancies goes beyond the Standards, the Standards require that you identify whether the equipment is near the *end* of its normal life expectancy. Although air conditioners have many components, the compressor is the heart of the unit. It is, by a wide margin, the most expensive component to replace on an air conditioner or heat pump.

Checking the Age

You can determine the age of the unit from the condenser unit data plate. **Preston's Guide** is helpful in this regard. Since compressors are often replaced, the home inspector who goes above and beyond will check on a unit that is more than 5 years old to determine whether the compressor has been replaced. This involves removing the cover from the outdoor condenser unit and checking the compressor

itself. There are only a few manufacturers of compressors and the date coding systems can be determined by contacting the manufacturers. Older systems with relatively young compressors are not uncommon. If you aren't going to go this far, when you see old air conditioners, you should allow for the possibility that the compressor has been replaced.

Life expectancies vary. Many people in the southern United States consider 8 to 10 years a typical compressor life expectancy. In more moderate climates, a 10- to 15-year life expectancy is typical, and in northern climates, life expectancies of 15 to 20 years may be appropriate.

Warranty

Many compressors are warranted for five years. This suggests a life expectancy range.

IMPLICATION

The life expectancy may have no implication with respect to existing performance. We may simply be anticipating breakdown. However, older compressors may suffer compression ratio decreases as valves wear. They may also be more expensive to operate because they draw more electricity. Some people recommend replacement of old compressors even if they are still operable, as a way to reduce operating costs through enhanced efficiency.

STRATEGY

The age can be determined from the data plate on the condenser or, as discussed, from the compressor itself. Interpretation of serial numbers is sometimes required to verify age. The compressors typically have a date tag riveted or glued onto the shell of the unit.

Speak with manufacturers and installers in your area to get a sense of the common life expectancies for air conditioning compressors in your climate. If you are going to include life expectancy comments in your reports, make it clear to the client you are dealing with a probability and not a certainty. Many compressors will fail before their normal life is up and others will go well beyond what is expected. You may also ask at what age it makes more sense to replace the whole system, rather than just the compressor.

12.12.1 Ductless Air Conditioning

So far our discussion has focused on air conditioning systems that employ ducts to move cooled air through the house and warm air back to the air conditioner. Let's look briefly at **ductless air conditioning**.

Alternative to Central Systems

Air conditioning is expensive to add to homes that do not have ducts. Ductless air conditioning is becoming a popular option because it is not disruptive to install and can provide several cooling zones. Ductless systems are available as air- or water-cooled units.

Split Systems

There are two common types of ductless systems; **split systems** and **single component systems**. Split systems have a condenser cabinet with a compressor, condenser coil and fan on the ground or on the roof, the same as any central split system. The evaporator coil and house air fan are inside the home, in the area to be cooled. There is a condensate collection and discharge system for the interior component. There are two refrigerant lines, often in a conduit, joining the outdoor condenser unit to the evaporator inside the home.

Compact

Split systems, also called **mini-splits**, are easy to install and require only a 3-inch diameter hole through the house wall. The indoor components can be wall or ceiling mounted and don't take up much space. Some are sold with remote controls so they can be mounted out of the way, high on walls or on ceilings.

<i>Quiet</i>	These have the advantage of an outdoor compressor (the noisiest part of an air conditioner) so the home is quieter. Some interior fans are multi-speed to minimize noise. There are also quieter condenser fans in some systems that operate at very low rpm (less than 900 rpm).
<i>Multi-zone Systems</i>	Split systems can be multi-zone, with one condenser unit serving up to four evaporators in different parts of the home.
<i>Large Capacity</i>	Split systems are available with cooling capacities up to 60,000 BTUS/hr (five tons), large enough for most homes.
<i>Single Component Systems</i>	Single component systems are also called through-wall or package systems . These are self-contained systems with the condenser, compressor and evaporator all in the same cabinet, installed in the wall of the room or area to be cooled. These units are common in motels and apartments. These single component systems are noisier than split systems because the compressor is in the wall. Some include electric elements for supplementary heating. Single component systems may be wired directly into the panel or may plug into a 240-volt receptacle.
<i>Inspection Issues</i>	The inspection procedures for ductless systems are similar to central systems. There is no distribution system to worry about, and there is often much less you can see, especially on single component systems. There is typically a filter access panel, but that is about all that is accessible. On ceiling and high-wall mounted units, the Standards suggest you don't have to open these panels since they are not within normal reach . Dirty air filters are a common problem with ductless systems, especially when the system is out of reach of the average person.
<i>Airflow Issues</i>	Ductless air conditioners can blow air up to 40 feet in an open area, but since there is no distribution system, even cooling in multiple rooms from a single system is unlikely. In small rooms, air can bounce off walls or furnishings and create short cycling and comfort problems. These systems are often located near the top of the stairwell in a two-story home in an effort to cool as much of the home as possible.
<i>Condensate Damage</i>	Condensate discharge systems are often on the building exterior, below the wall-mounted evaporator. Discoloration or damage to the wall is a possibility if the condensate is allowed to run down the wall surface.

12.13 AN OVERVIEW OF HEAT PUMPS

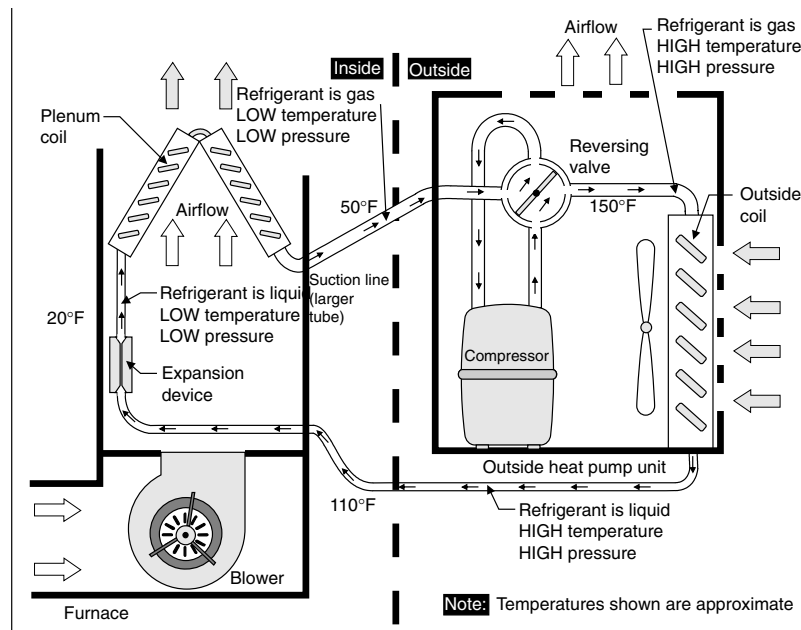
The people who designed, built and sold air conditioning equipment knew that it was fairly expensive. In many parts of North America, the equipment sits idle much of the year. If they could use the same equipment that cools the house in the summer to heat it in the winter, wouldn't that be great! They found with some modifications, they could do just that. The result is the heat pump (Figure 12.18).

Inspecting central air conditioning is hard enough. Inspecting heat pumps is even more challenging. Not only is it an air conditioning system that sometimes acts as a heating system, there is often a back-up heating system to inspect as well. Doesn't seem fair, does it?

A discussion of the mechanics of heat pumps is beyond the scope of this book, but we will get a quick overview and then look at typical conditions of heat pumps.

At first glance, it can be difficult to differentiate heat pumps from air conditioners, especially during the summer months. The following are some of the ways to tell.

1. Look at the data plate. It may say **Heat Pump** on it, or the model number may start with **HP**. If it does not, you can jot down the manufacturer's name and model number and contact the manufacturer.

FIGURE 12.18 Heat Pump Schematic—Summer Mode

2. If the thermostat has an **Emergency Heat** setting, you can be sure it's a heat pump.
3. If you take the cover off the thermostat (beyond the Standards) and find that it's a two-stage thermostat, this indicates a heat pump. (You must know what you're looking at here.)
4. If both refrigerant lines are insulated, it's a heat pump.
5. If you open the condenser cabinet and find a reversing valve, it's a heat pump.
6. If you find two expansion devices with bypasses, it's a heat pump.
7. If the compressor is indoors and it's an air to air system, it's a heat pump.
8. If there is an outdoor thermostat connected to the control wiring, it's probably a heat pump.
9. If it's winter and the unit is operating, it's a heat pump.

Is It Still Being Used as a Heat Pump?

In many northern climates, heat pumps were often installed with the encouragement of local utilities. For various reasons, they were found to be unsatisfactory. Many have been disconnected as heat pumps and are used only as air conditioners. You won't be able to determine this visually, but your client may want to ask the current owner whether the heat pump is still used for heating. In many cases, it is cheaper to have the back-up heat supply all of the heating, especially if the back-up is a high-efficiency gas furnace. Where the available back-up heat is electricity, heat pumps are more cost effective and more likely to be in service, even in colder climates.

If you're looking at the house in the winter, and the system is operating on the **Emergency Heat** setting, chances are the heat pump is not operative.

12.14 HEAT PUMP CONDITIONS

All of the problems associated with air conditioning components may also be experienced with heat pumps. The following additional problems are common on heat pumps:

1. Oversized for cooling and/or undersized for heating
2. Heat pump inoperative in heating or cooling mode
3. Poor outdoor coil location, including
 - a. under the drip line of the roof
 - b. where snow drifts accumulate
 - c. where air re-circulates and is trapped in an enclosed area, or
 - d. where the outdoor coil is exposed to the prevailing wind
4. Outdoor coil is iced up
5. Airflow problems
6. No back-up heat, or back-up heat does not work
7. Old

12.14.1 Oversized for Cooling and Undersized for Heating

Almost all heat pumps in northern climates are undersized for heating. Their sizing should be determined by the cooling load. There is a temptation to oversize for cooling so that the heating load can be satisfied by the heat pump. This is usually poor practice.

IMPLICATIONS

Inappropriate sizing may mean that auxiliary heat is needed in the winter. In the summer, the cooling may produce a damp unpleasant indoor climate and the system may short cycle.

STRATEGY

Since we're not going to do heat loss or heat gain calculations, your estimates of appropriate size will be very rough. Don't go out on a limb and condemn a unit as being undersized or oversized. You're wiser to describe it as suspect, and recommend further investigation. In northern climates, look for auxiliary heat to help the heat pump when it's cold.

12.14.2 Inoperative in Heating or Cooling Mode

IMPLICATIONS

If the system is inoperative in heating mode, there may be no heat, or there may be complete dependence on the auxiliary heat. If the system doesn't work in the cooling mode, there will be no air conditioning and may be no heat.

STRATEGY

Turn the thermostat up to see if the heat pump responds (assuming the outdoor temperature is below 65°F). Check that the heat pump is producing the heat, not the auxiliary heat. If the system is inoperative, a service person should be contacted, since troubleshooting heat pumps is well beyond our scope.

Depending on the outdoor temperature, the type of back-up heat, and the set-up of the system, the heat pump may be operational but locked out.

Locked Out

If the system is set to the cooling mode and the outdoor temperature is above 65°F, the system should respond to lowering the thermostat. If it doesn't respond or doesn't deliver cool air, recommend further investigation. Let your client know that the system may not deliver any heat either, although you can't test that.

12.14.3 Poor Outdoor Coil Location

IMPLICATION

Inefficient operation, decreased comfort and increased costs will result if the outdoor coil is not well located.

STRATEGY

Watch for outdoor units—

- with restricted intake or discharge air flow
- within 6 feet of a clothes dryer, water heater vent or high efficiency furnace vent
- under the drip line of the roof
- below snow drift depth or with no provision for condensate to drain away freely

Some say the best location for a heat pump outside is on the east side of the house, but practical issues are probably more important.

We added furnace discharge vents to the list here. These vents are not an issue for air conditioning systems because the air conditioner and furnace would never operate at the same time. It can be an issue with heat pumps because heat pumps and furnaces may operate at the same time. The moisture in the furnace exhaust may ice up the heat pump coil quickly.

Where snow tends to accumulate, the heat pump should be on a frame or stand several inches above the ground. This helps to prevent snow from blocking the air movement across the coil. There should be 1 to 3 feet of clearance for the intake and 4 to 6 feet for the discharge side.

Snow Accumulation Areas

12.14.4 Coil Iced Up

IMPLICATION

Poor heating or no heat will result from an iced up coil.

STRATEGY

Look for frost or ice on the outdoor coil.

12.14.5 Airflow Problems in House

IMPLICATIONS

Poor heating and cooling can result from airflow problems.

STRATEGY

Perform your tests of the supply registers and return grilles as we have discussed previously.

Watch for supply registers and return grills that favor either heating or cooling. For example, ceiling supply registers are fine for cooling but are not very good for heating.

Low level returns are better for heating than for cooling. Ideal systems have high and low supply registers and return grilles. These are very rare. The best compromise for both heating and cooling is floor registers near outside walls below windows. Return grilles can be high and low with a damper on the low return grille. The damper is open in the heating season for low level return and closed in the cooling season allowing high level return.

Poor Register and Grille Location

Ducts in Unconditioned Spaces

The more of the duct system that passes through unconditioned spaces (attics and crawlspaces), the worse the situation is. Comfort suffers and costs increase. Insulating and sealing ducts helps, but is never 100 percent effective. Warn your clients if there is much ductwork in unconditioned spaces.

12.14.6 Back-Up Heat Problems

IMPLICATION
STRATEGY

The back-up heat may be inactive or ineffective. The result is often inadequate heat in cold weather.

When looking at a heat pump, determine whether back-up heat is needed. You should know this from local installation practices. You can also approximate this by knowing the rough heat loss from the house and determining the heating capacity of the heat pump.

Back-Up Heat

If back-up heat is needed, has it been provided? What type is it? If the back-up heat is electric, it can operate in conjunction with the heat pump. If the back-up heat is gas or oil, the heat pump has to be shut off when the gas or oil burner comes on.

Testing Back-Up Heat

If the back-up heat is electric, it may be several staged elements that come on one at a time as required. You should, as part of your inspection, make sure that auxiliary heat is provided. You should also make sure that it works. Switching to the Emergency Heat setting, and using an ampmeter to measure the current draw through the auxiliary heating circuit is one way to accomplish this. If it is a staged system, you should make sure that each of the auxiliary electric elements comes on appropriately.

Temperature Limitations

You should avoid testing a heat pump in the heating mode above 65°F. Do not run a heat pump in the cooling mode below 65°F.

Back-Up Heat on Constantly

The electric back-up heat should only come on when the heat pump can't deliver enough heat. If the back-up heat is on with the heat pump when the temperature is mild (above 40°F), there may be a wiring or control setting problem. Recommend further investigation.

Five to Seven Minutes Between Cycles

Heat pumps, just like air conditioners, have a built-in time delay to prevent short on/off cycles which create a large pressure for the compressor to try to start up against. This can damage the compressor. Any time the heat pump is shut off, it probably won't recycle for five to seven minutes. A few manufacturers go even longer before allowing the system to re-start.

12.14.7 Old

Heat pump compressors last 8 to 15 years, typically. Use the methods discussed in the Section 12.13 to check the age of the heat pump and compressor.

REVIEW QUESTIONS

1. List ten items that affect the amount of air conditioning needed in a home.
2. How many square feet can one ton of air conditioning cool in Florida?
3. How many square feet can one ton of air conditioning cool in Michigan?
4. Oversized distribution ductwork is a common problem with central air conditioning.
True False
5. An undersized air conditioner is better than an oversized one.
True False
6. The typical temperature drop from outdoors to indoors with a properly operating air conditioning system would be about 15°F.
True False
7. What kind of temperature drop would you expect to find in the house air as it passes over the evaporator coil?
8. A compressor can be thought of as a pump.
True False
9. Compressors are typically located indoors on an air-to-air split system air conditioning system.
True False
10. What is the function of a sump or crankcase heater?
11. List eight common compressor problems.
12. The condenser coil in a split system is typically located _____.
13. The evaporator coil in a split system is typically located _____.
14. The condenser fan in a split system is located _____.
15. List four condenser problems.
16. List three common configurations of evaporator coils.
17. Evaporator coils should be upstream of furnace heat exchangers.
True False
18. List six common evaporator problems.
19. In one sentence, describe the function of a capillary tube.
20. In one sentence, describe the function of a thermostatic valve.
21. List four common expansion device problems.
22. List five possible causes of leaking condensate drain pans.
23. When is an auxiliary condensate drain pan needed?
24. What are condensate drain lines usually made of?
25. Do condensate drain lines ever have a trap?
26. If so, what is its purpose?

27. List two acceptable discharge points for condensate drain lines and one generally unacceptable discharge point.
28. List three common condensate pump problems.
29. Refrigerant lines are usually made of _____.
30. The larger line contains a liquid.
True False
31. The smaller line is called the suction line.
True False
32. The larger line is called the return line.
True False
33. List five common refrigerant line problems.
34. How would you normally identify a leak in a refrigerant line visually?
35. Which line should be cold when the system is operating?
36. Which line should be warm when the system is operating?
37. Which line should be insulated?
38. Where is the condenser fan in a split system located?
39. What is the typical inlet and outlet temperature of air across the condenser fan?
40. List four common condenser fan problems
41. Where is the evaporator fan typically located in a split system?
42. The amount of air that must cross an evaporator coil in a heating climate is less than the amount of air that must cross a conventional furnace heat exchanger coil.
True False
43. Oversized ducts are a common problem.
True False
44. High level returns are more appropriate for heating than cooling.
True False
45. Ducts passing through unconditioned spaces such as attics or crawlspaces should be _____.
46. List seven common problems with thermostats.
47. How long do air conditioning compressors typically last in your area?
48. What is the typical warranty period for a compressor?
49. Heat pumps are generally sized based on the
 - a. heating load
 - b. cooling load
 - c. other

50. The implication of an oversized heat pump during the summer months is:
51. You should not operate the heat pump in the heating mode when the outdoor temperature is above _____.
52. You should not operate the heat pump in the cooling mode when the outdoor temperature is below _____.
53. List at least eight clues that would suggest that you are looking at a heat pump rather than a central air conditioning system.

